



US012408401B2

(12) **United States Patent**
Kinoshita

(10) **Patent No.:** **US 12,408,401 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SILICON CARBIDE SEMICONDUCTOR
DEVICE AND METHOD OF
MANUFACTURING SILICON CARBIDE
SEMICONDUCTOR DEVICE**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 385 days.

(21) Appl. No.: **17/682,589**

(22) Filed: **Feb. 28, 2022**

(65) **Prior Publication Data**
US 2022/0344475 A1 Oct. 27, 2022

(30) **Foreign Application Priority Data**
Apr. 27, 2021 (JP) 2021-074572

(51) **Int. Cl.**
H10D 62/832 (2025.01)
H10D 62/10 (2025.01)

(52) **U.S. Cl.**
CPC **H10D 62/8325** (2025.01); **H10D 62/105**
(2025.01)

(58) **Field of Classification Search**
CPC H01L 29/1608; H01L 29/0615; H01L
29/0623; H01L 29/0638; H01L 29/1095;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,691,553 A * 11/1997 Mori H01L 29/408
257/490
5,994,754 A * 11/1999 Hayashi H01L 29/0615
257/490

(Continued)

FOREIGN PATENT DOCUMENTS

JP H08-88346 A 4/1996
JP 2002368215 A 12/2002

(Continued)

OTHER PUBLICATIONS

Extended European Search Report in the counterpart European
Patent Application No. 22163968.5 dated Oct. 10, 2022.

(Continued)

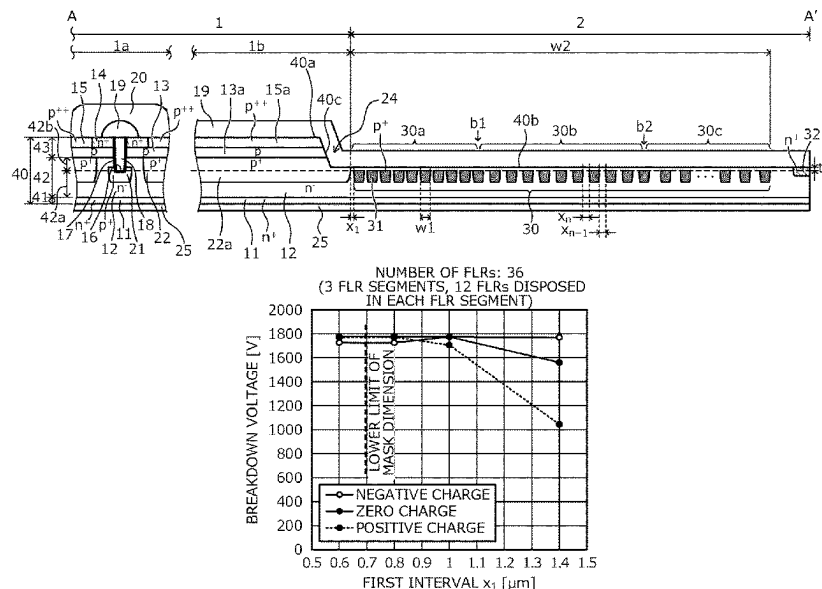
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(57) **ABSTRACT**

In an edge termination region, a FLR structure configured by
FLRs having a floating potential and surrounding concentrically a periphery of an active region is provided. The FLR
structure is divided into at least two FLR segments with a
predetermined FLR as a boundary. An n-th interval between
an adjacent two of the FLRs is wider than a first interval
between a p⁺-type extension portion and the FLR closest to
a chip center (n=2 to total number of the FLRs). The n-th
interval between an adjacent two of the FLRs increases in
arithmetic progression the closer the adjacent two are to a
chip end, the n-th interval increasing in arithmetic progres-
sion by a corresponding one of constant increase increments
respectively corresponding to the FLR segments; the closer
a FLR segment is to the chip end, the wider is the constant
increase increment corresponding thereto.

13 Claims, 10 Drawing Sheets



(58) **Field of Classification Search**

CPC H01L 29/0619; H01L 29/0661; H01L
29/66068; H01L 29/7811; H01L 29/7813;
H01L 21/265; H01L 29/06; H10D
62/8325; H10D 62/105; H10D 30/0245;
H10F 71/139

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

11,527,660 B2 * 12/2022 Hoshi H01L 29/7813
2003/0042549 A1 3/2003 Fujihira et al.
2003/0218220 A1 * 11/2003 Takahashi H10D 12/441
257/E29.198
2009/0200559 A1 * 8/2009 Suzuki H01L 29/1095
257/77
2011/0291241 A1 * 12/2011 Yoshikawa H10D 12/441
257/544
2014/0197422 A1 * 7/2014 Wada H01L 29/0619
257/77
2014/0327019 A1 * 11/2014 Kinoshita H01L 29/41
257/77
2015/0084124 A1 * 3/2015 Saito H10D 62/106
257/330
2015/0279931 A1 * 10/2015 Chen H10D 12/441
257/495
2016/0049463 A1 * 2/2016 Buchholz H10D 8/60
257/493
2017/0012122 A1 * 1/2017 Takaya H01L 29/66068

2017/0179235 A1 * 6/2017 Kinoshita H01L 29/0615
2018/0019301 A1 * 1/2018 Yasuda H01L 29/872
2018/0040688 A1 2/2018 Kobayashi et al.
2018/0166530 A1 6/2018 Lichtenwalner et al.
2019/0131412 A1 * 5/2019 Ozaki H10D 62/105
2019/0341308 A1 * 11/2019 Urakami H01L 29/7397
2019/0386095 A1 * 12/2019 Takeuchi H01L 29/0878
2020/0006409 A1 * 1/2020 Bakowski Holtryd
G01N 23/046
2020/0227549 A1 * 7/2020 Noborio H01L 29/66734
2020/0328273 A1 * 10/2020 Sakata H10D 30/665
2021/0083044 A1 * 3/2021 Yoshikawa H01L 29/8611
2022/0093729 A1 * 3/2022 Kanie H10D 62/8325
2022/0199824 A1 * 6/2022 Kinoshita H01L 29/1095
2022/0344475 A1 * 10/2022 Kinoshita H01L 29/0661
2023/0062475 A1 * 3/2023 Kumada H01L 29/66068

FOREIGN PATENT DOCUMENTS

JP 2008010506 A 1/2008
JP 2011-101036 A 5/2011
JP 2014-232838 A 12/2014
JP 2018022851 A 2/2018
JP 2019054087 A 4/2019

OTHER PUBLICATIONS

Japanese Office Action dated Jan. 14, 2025, in the counterpart
Japanese Patent Application No. 2021-074572.

* cited by examiner

FIG. 1

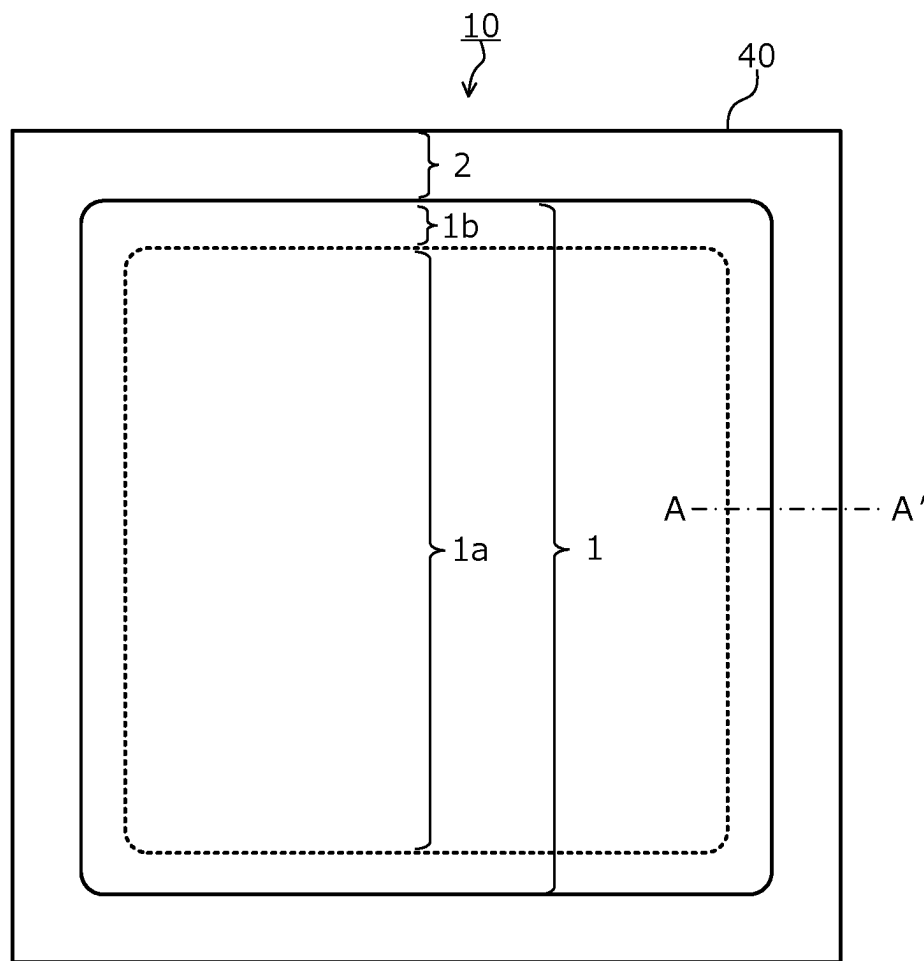


FIG. 2

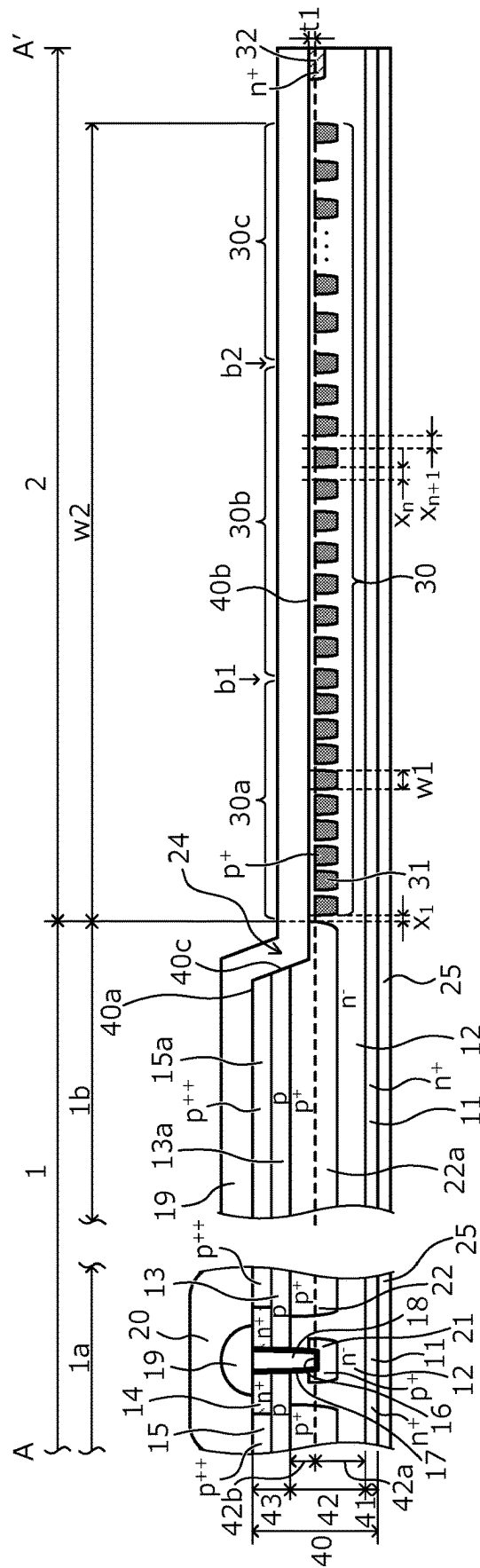


FIG.3

		FLR SEGMENT 30a										FLR SEGMENT 30b				
No.		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
INTERVAL x _m (μm)		1	1.05	1.1	1.15	1.2	1.25	1.3	1.35	1.4	1.45	1.53	1.61	1.69	1.77	1.85

		FLR SEGMENT 30b										FLR SEGMENT 30c									
No.		16	17	18	19	20	21	22	23	24	25	26	27	28	29	30					
INTERVAL x _m (μm)		1.93	2.01	2.09	2.17	2.25	2.37	2.49	2.61	2.73	2.85	2.97	3.09	3.21	3.33	3.45					

FIG. 4

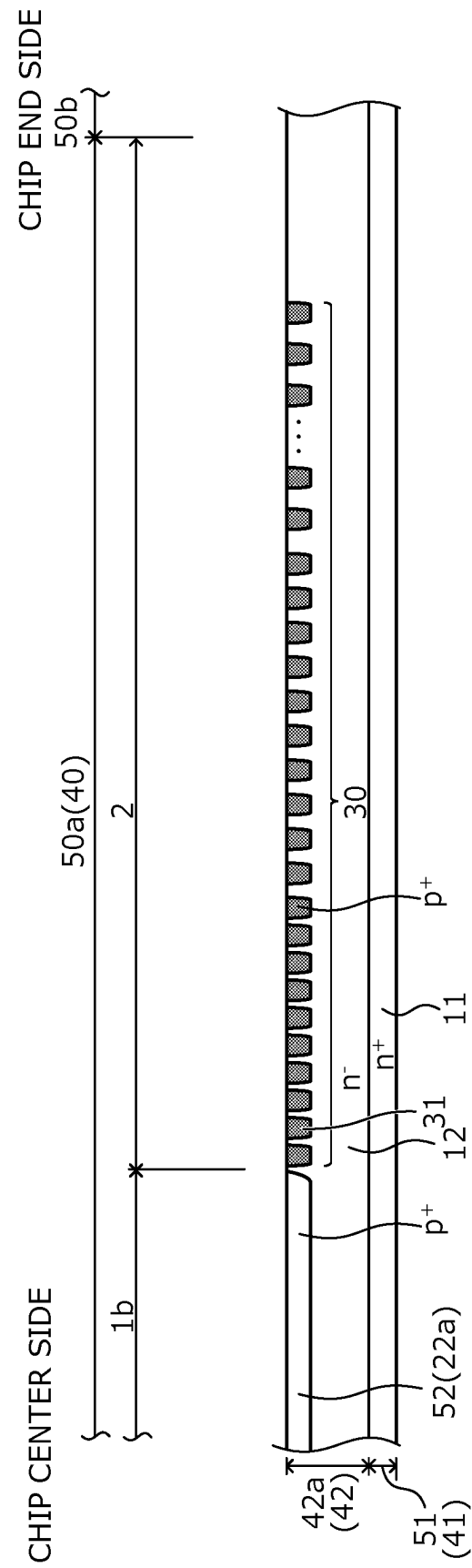


FIG.5

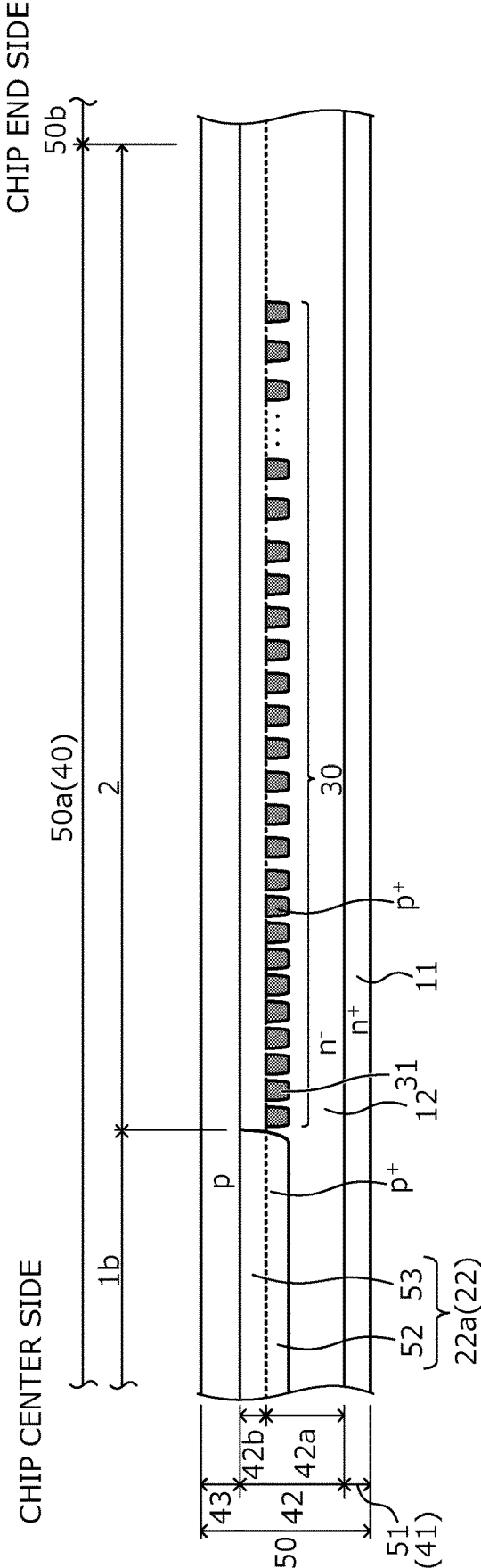


FIG. 6

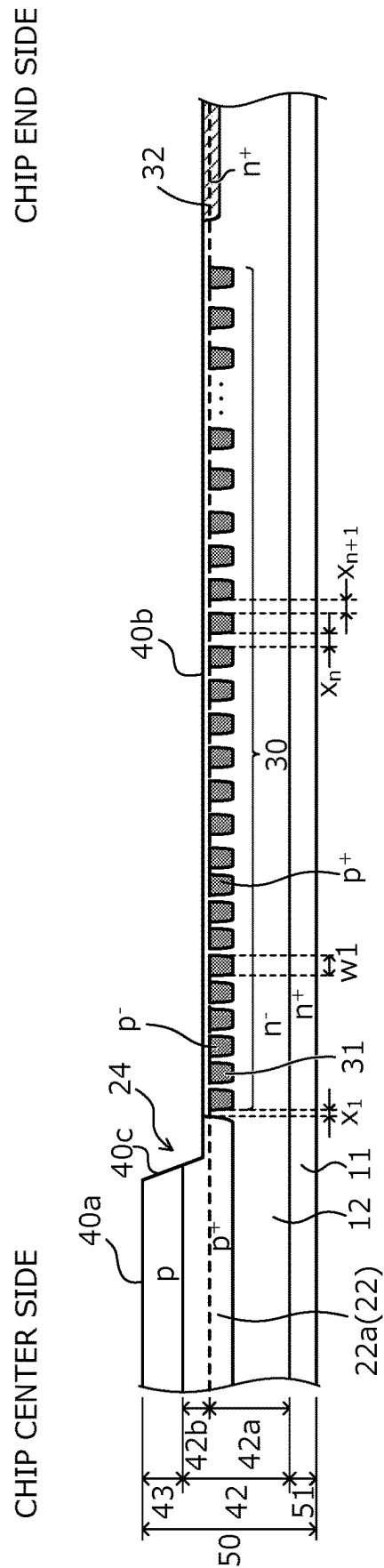


FIG.7 RELATED ART

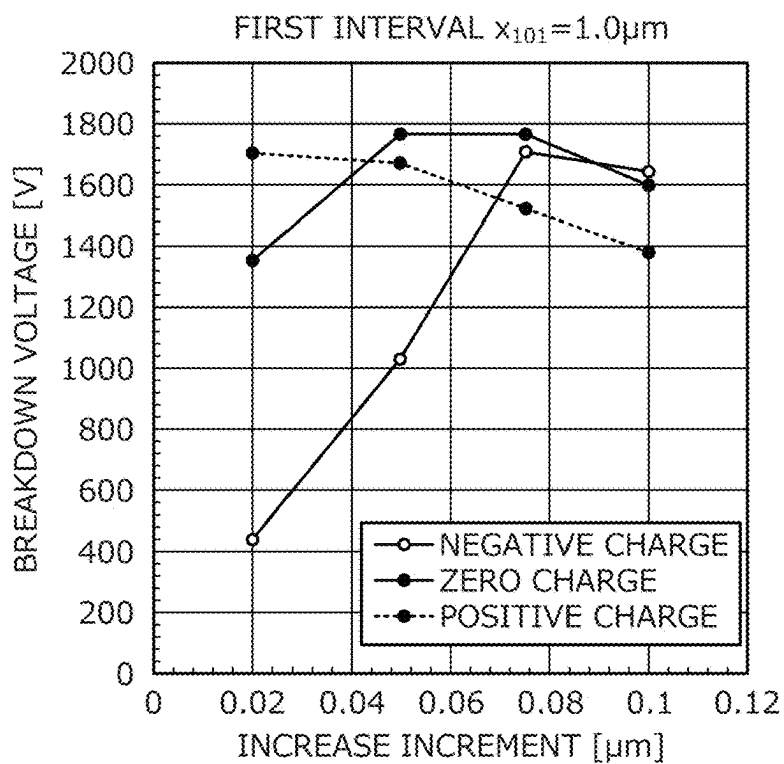


FIG.8 RELATED ART

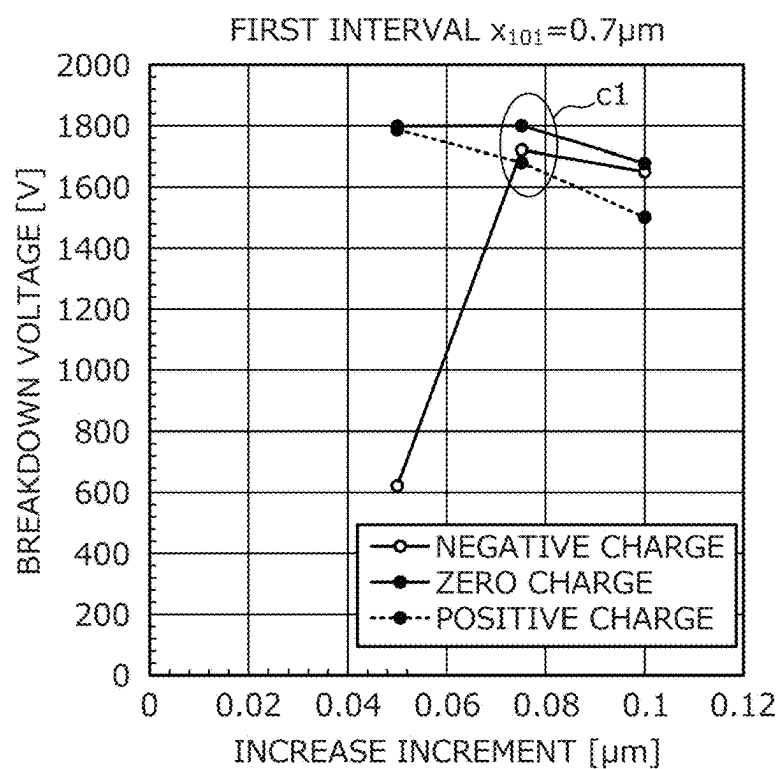


FIG.9 RELATED ART

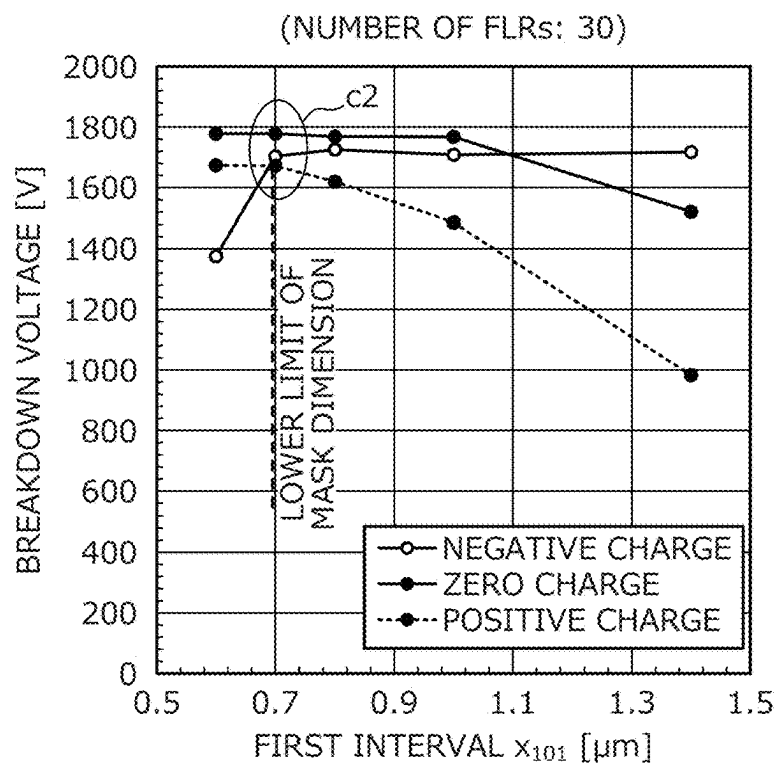


FIG.10 RELATED ART

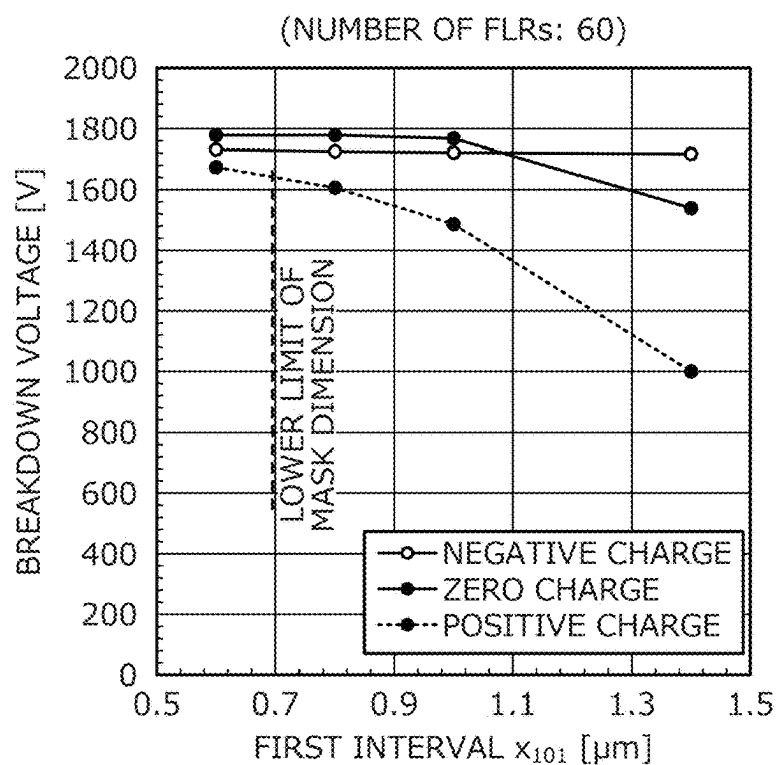


FIG. 11

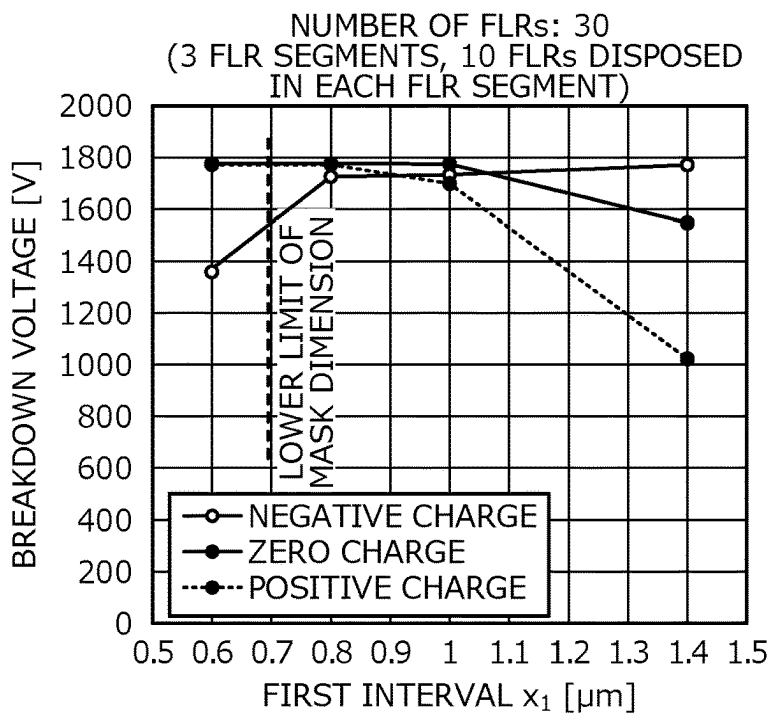


FIG. 12

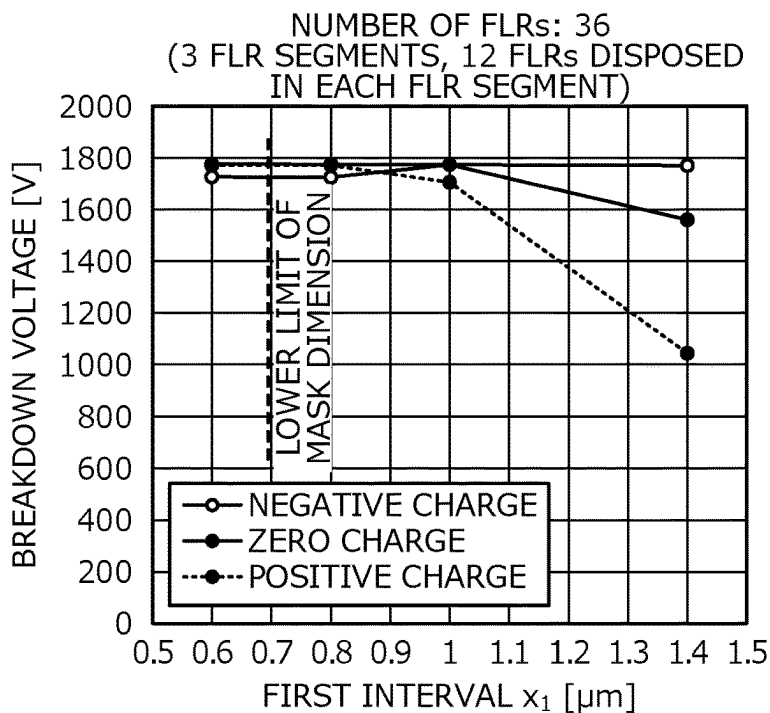
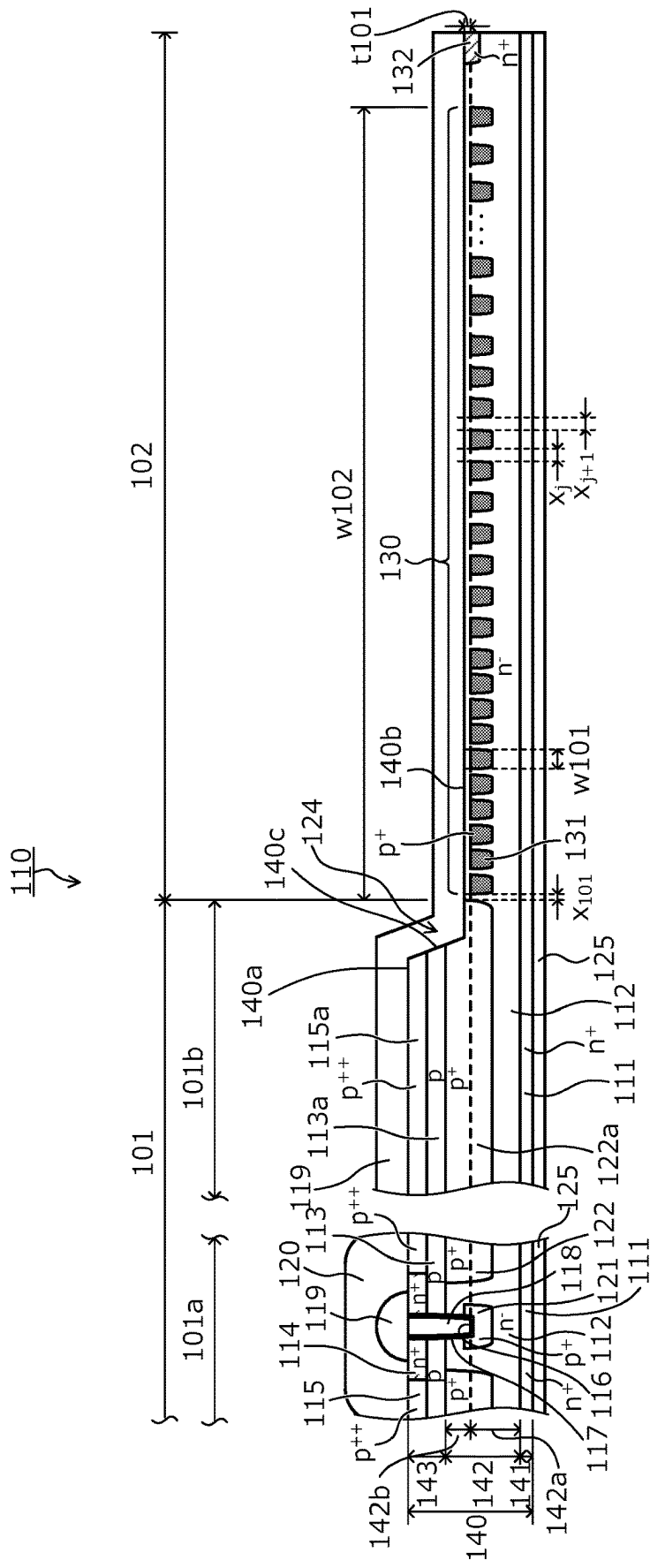


FIG. 13
RELATED ART



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SILICON CARBIDE SEMICONDUCTOR DEVICE AND METHOD OF MANUFACTURING SILICON CARBIDE SEMICONDUCTOR DEVICE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2021-074572, filed on Apr. 27, 2021, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

Embodiments of the invention relate to a silicon carbide semiconductor device and method of manufacturing a silicon carbide semiconductor device.

2. Description of the Related Art

A voltage withstanding structure of a power semiconductor device is configured by multiple p-type regions selectively provided in surface regions of an n-type drift region exposed at a front surface of a semiconductor substrate, in an edge termination region between an active region and an end of a semiconductor substrate (semiconductor chip). In an instance in which a semiconductor material of the power semiconductor device is silicon carbide (SiC) for which a critical electric field strength is at least 10 times a critical electric field strength of silicon (Si), a multizone junction termination extension (JTE) structure, a spatially modulated JTE structure, a field limiting ring (FLR) structure are mainly disposed as the voltage withstanding structure.

The multizone JTE structure is a structure in which at least three p-type regions (hereinafter, JTE regions) are disposed adjacent to one another concentrically surrounding a periphery of the active region so as to be in descending order of impurity concentration from a chip center side (active region (center portion of the semiconductor chip: chip center) side) to a chip outer side (semiconductor substrate end (chip end)). The electric field strength tends to decrease with increasing distance from the active region in a direction to the chip end. Therefore, in the multizone JTE structure, according to the tendency of the electric field strength distribution, the impurity concentrations of corresponding JTE regions disposed relatively closer to the chip outer side are lower than the impurity concentrations of JTE regions disposed relatively closer to the active region, whereby a predetermined breakdown voltage is stably ensured.

The spatially modulated JTE structure is an improved structure of the JTE structure and between JTE regions adjacent to one another (in an instance of a configuration having only one JTE region, between the one JTE region and an n-type drift region on an outer side thereof facing the chip end), a spatially modulated region having an intermediate impurity concentration of these two regions and having a spatially equivalent impurity concentration distribution is disposed, the spatially modulated JTE structure being a structure in which an impurity concentration distribution of the JTE structure overall gradually decreases in a direction to the chip end. The spatially modulated region is formed by repeatedly disposing two small regions in a predetermined pattern, two small regions respectively having the same

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impurity concentration as regions adjacent thereto. The spatial impurity concentration distribution of an entire area of the spatially modulated region is determined by widths and an impurity concentration ratio of the two small regions.

The spatially modulated JTE structure may stably ensure the predetermined breakdown voltage compared to a general JTE structure without the spatially modulated region. The FLR structure is a structure in which multiple p-type regions (FLRs) having the same impurity concentration are disposed apart from one another concentrically surrounding a periphery of the active region. In the FLR structure, electric field is distributed by the multiple FLRs disposed apart from one another, thereby ensuring the predetermined breakdown voltage. In this manner, a predetermined voltage withstanding structure is disposed in the edge termination region, thereby mitigating or distributing the electric field of the edge termination region, whereby the breakdown voltage of the edge termination region is enhanced and thus the breakdown voltage of the semiconductor device overall is enhanced. A structure of a conventional silicon carbide semiconductor device is described.

FIG. 13 is a cross-sectional view depicting the structure of the conventional silicon carbide semiconductor device. A conventional silicon carbide semiconductor device 110 depicted in FIG. 13 is a vertical metal oxide semiconductor field effect transistor (MOSFET) with a trench structure having insulated gates (MOS gates) having a three-layer structure including a metal, an oxide film, and a semiconductor, and a FLR structure 130 provided in an edge termination region 102 of a semiconductor substrate (semiconductor chip) 140 containing silicon carbide. The edge termination region 102 surrounds a periphery of an active region 101 through which a main current of the MOSFET flows.

In the semiconductor substrate 140, silicon carbide layers 142, 143 constituting an n⁻-type drift region 112 and a p-type base region 113 are sequentially grown epitaxially on an n⁺-type starting substrate 111 containing silicon carbide. A portion of the p-type silicon carbide layer 143 in the edge termination region 102 is removed by etching, thereby forming a drop 124 at a front surface of the semiconductor substrate 140. The front surface of the semiconductor substrate 140 is recessed toward a drain electrode 125, at a second surface portion 140b closer to the chip outer side (end side of the semiconductor substrate 140) than is a first surface portion 140a of a chip center (center of the semiconductor substrate 140) side, with the drop 124 as a boundary.

Due to the drop 124, the p-type silicon carbide layer 143 is left in a mesa shape in a center of the front surface (main surface having the p-type silicon carbide layer 143) of the semiconductor substrate 140. The first and the second surface portions 140a, 140b of the front surface of the semiconductor substrate 140 are formed by the p-type silicon carbide layer 143 and the n-type the silicon carbide layer 142, respectively. In a center portion 101a of the active region 101, MOS gates constituted by trenches 116, gate insulating films 117, and gate electrodes 118, and the p-type base region 113, n⁺-type source regions 114, and p⁺⁺-type contact regions 115 of the unit cells of the MOSFET are provided in a front side of the semiconductor substrate 140.

In a peripheral portion 101b of the active region 101, in a surface region at the first surface portion 140a of the front surface of the semiconductor substrate 140, a p⁺-type extension portion 122a, a p-type base extension portion 113a, and a p⁺⁺-type contact extension portion 115a are selectively provided. The p⁺-type extension portion 122a, the p-type

base extension portion 113a, and the p⁺⁺-type contact extension portion 115a are, respectively, extended portions of an outermost p⁺-type region 122, the p-type base region 113, and the p⁺⁺-type contact region 115 that is outermost, these regions configure a unit cell of the center portion 101a of the active region 101 and surround a periphery of the center portion 101a of the active region 101.

The p⁺-type extension portion 122a, the p-type base extension portion 113a, and the p⁺⁺-type contact extension portion 115a extend toward the chip outer side and reach a third surface portion (mesa edge of the drop) 140c connecting the first surface portion 140a and the second surface portion 140b of the front surface of the semiconductor substrate 140. The p⁺-type extension portion 122a extends closer to the chip outer side than does the drop 124 and is exposed at the second surface portion 140b of the front surface of the semiconductor substrate 140. Reference numerals 121 and 122 are p⁺-type regions that mitigate electric field applied to the gate insulating films 117 at bottoms of the trenches 116. Reference numerals 119, 120, 125, 132 are, respectively, an interlayer insulating film, a source electrode, a drain electrode, and an n⁺-type channel stopper region.

In the edge termination region 102, multiple p-type regions (FLRs: hatched portion) 131 of a floating potential and configuring the FLR structure 130 are selectively provided in the n⁻-type silicon carbide layer 142, in a surface region thereof at the second surface portion 140b of the front surface of the semiconductor substrate 140. The FLRs 131 are provided apart from one another, concentrically surrounding a periphery of the active region 101, at positions apart from the p⁺-type extension portion 122a and closer to the chip outer side than is the p⁺-type extension portion 122a. All the FLRs 131 face the p⁺-type extension portion 122a in a normal direction (direction from the chip center side to the chip outer side). All the FLRs 131 are surrounded at peripheries thereof by the n⁻-type drift region 112.

A first interval between the p⁺-type extension portion 122a and the FLR 131 closest to the chip center is assumed as x_{101} while an i-th interval between an i-th FLR 131 and an (i-1)-th FLR 131 adjacent thereto on a chip center side thereof and subsequent intervals in a direction to the chip end are assumed as $x_j, x_{j+1}, \dots$, respectively (where, $i=2$ to a total number of the FLRs 131, $j=i+100$). An i-th interval x_j between an adjacent two of the FLRs 131 is wider than the first interval x_{101} between the p⁺-type extension portion 122a and the FLR 131 closest to the chip center, and the i-th interval x_j between an adjacent two of the FLRs increases in arithmetic progression by a constant increase increment (amount in the normal direction) ($x_{j+1}-x_j=\text{constant}$), the closer the adjacent two of the FLRs are to the chip end. All the FLRs 131 have a same impurity concentration and a same width w101.

As a conventional silicon carbide semiconductor device, a device having a FLR structure in which an interval between an adjacent two of the FLRs increases in arithmetic progression by a constant increase increment, the closer the adjacent two of the FLRs are to the chip end has been proposed (for example, refer to Japanese Laid-Open Patent Publication No. 2011-101036 and Japanese Laid-Open Patent Publication No. H8-088346). In Japanese Laid-Open Patent Publication No. 2011-101036, due to a p-type RESURF region between the active region and the FLR structure, when a drain-source voltage becomes a high voltage, electric field concentrates near the FLR structure, not directly beneath the p-type RESURF region. In Japanese

Laid-Open Patent Publication No. H8-088346, between adjacent FLRs, blocked voltage is stabilized by a field plate (FP) provided on an n⁻-type drift region via an insulating layer.

Further, as a conventional silicon carbide semiconductor device, a device having, in an edge termination region, a FLR structure configured by multiple FLRs concentrically surrounding a periphery of an active region and disposed at equal intervals, and an n-type region provided between a second portion of a front surface of a semiconductor substrate and the multiple FLRs has been proposed (for example, refer to Japanese Laid-Open Patent Publication No. 2014-232838). In Japanese Laid-Open Patent Publication No. 2014-232838, due to the n-type region between the second portion of the front surface of the semiconductor substrate and the multiple FLRs, the FLRs are disposed at positions apart from the second portion of the front surface of the semiconductor substrate, whereby variation of a dose amount of the FLRs is reduced and variation of the breakdown voltage of the silicon carbide semiconductor device is reduced.

SUMMARY OF THE INVENTION

According to an embodiment of the invention, a silicon carbide semiconductor device includes a semiconductor substrate containing silicon carbide and having an active region and a termination region surrounding a periphery of the active region, the semiconductor substrate having a first main surface and a second main surface opposite to each other; a first semiconductor region of a first conductivity type, provided in the semiconductor substrate, spanning the active region and the termination region; a second semiconductor region of a second conductivity type, provided between the first main surface of the semiconductor substrate and the first semiconductor region in the active region; a device element structure having a pn junction between the first semiconductor region and the second semiconductor region, a current that energizes the pn junction flows in the device element structure; a second-conductivity-type peripheral region surrounding a periphery of the device element structure, provided between the first main surface of the semiconductor substrate and the first semiconductor region, between the device element structure and the termination region; a first electrode provided on the first main surface of the semiconductor substrate, electrically connected to the second semiconductor region and the second-conductivity-type peripheral region; a second electrode provided on the second main surface of the semiconductor substrate, electrically connected to the first semiconductor region; and a plurality of field limiting rings (FLRs) of the second conductivity type, the FLRs having a floating potential and configuring a FLR structure, the FLRs being provided apart from one another with respective intervals therebetween, each interval being a shortest distance between an adjacent two of the FLRs that are adjacent to each other, and each FLR concentrically surrounding the periphery of the active region, the plurality of FLRs being provided between the first main surface of the semiconductor substrate and the first semiconductor region in the termination region, the plurality of FLRs each facing a surface of the second-conductivity-type peripheral region, the surface of the second-conductivity-type peripheral region facing the termination region in the direction parallel to the first main surface of the semiconductor substrate. The plurality of FLRs includes at least one boundary FLR, and the FLR structure is divided into at least two FLR segments

with the at least one boundary FLR as a boundary. An interval between any adjacent two of the FLRs that are adjacent to each other is wider than an interval between the second-conductivity-type peripheral region and a first FLR of the plurality of FLRs closest to the second-conductivity-type peripheral region of the plurality of FLRs. Within a same FLR segment, an interval of an adjacent two of the FLRs increases as the adjacent two of the FLRs approaches closer to an end of the semiconductor substrate by a constant increment in arithmetic progression. Each constant increment of each FLR segment among the at least two FLR segments increases as said each FLR segment approaches closer to the end of the semiconductor substrate.

Objects, features, and advantages of the present invention are specifically set forth in or will become apparent from the following detailed description of the invention when read in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a plan view depicting a layout when a semiconductor device according to an embodiment is viewed from a front side of a semiconductor substrate.

FIG. 2 is a cross-sectional view of a structure along cutting line A-A' in FIG. 1.

FIG. 3 is a table depicting examples of dimensions of intervals between adjacent FLRs in FIG. 2.

FIG. 4 is a cross-sectional views of a state of a silicon carbide semiconductor device according to the embodiment during manufacture.

FIG. 5 is a cross-sectional views of a state of the silicon carbide semiconductor device according to the embodiment during manufacture.

FIG. 6 is a cross-sectional views of a state of the silicon carbide semiconductor device according to the embodiment during manufacture.

FIG. 7 is a characteristics diagram depicting results of simulation of breakdown voltage characteristics of a conventional example.

FIG. 8 is a characteristics diagram depicting results of simulation of breakdown voltage characteristics of the conventional example.

FIG. 9 is a characteristics diagram depicting results of simulation of breakdown voltage characteristics of the conventional example.

FIG. 10 is a characteristics diagram depicting results of simulation of breakdown voltage characteristics of the conventional example.

FIG. 11 is a characteristics diagram depicting results of simulation of breakdown voltage characteristics of a first examined example.

FIG. 12 is a characteristics diagram depicting results of simulation of breakdown voltage characteristics of a second examined examples.

FIG. 13 is a cross-sectional view depicting a structure of a conventional silicon carbide semiconductor device.

DETAILED DESCRIPTION OF THE INVENTION

First, problems associated with the conventional techniques are discussed. In a general JTE structure, to adjust the impurity concentration distribution according to tendency of the electric field strength distribution, the number of times ion implantation has to be performed is equal to the number of disposed JTE regions (p-type regions) having different impurity concentrations, whereby the number of processes

increases, leading to increases in cost. In the spatially modulated JTE structure, while the number of ion implantation is reduced, adjustment of the impurity concentration distribution according to the tendency of the electric field strength distribution is necessary and therefore, the number of processes increases, leading to increases in cost.

In the conventional FLR structure 130 (refer to FIG. 13), all the FLRs 131 have the same impurity concentration and therefore, ion implantation suffices to be performed once, thereby reducing the number of processes and suppressing manufacturing cost nonetheless to stably ensure the predetermined breakdown voltage, a large area (length of the edge termination region 102) is necessary. Therefore, in an instance in which silicon carbide for which the wafer cost is high is used as a semiconductor material, increases in material cost become a major factor in increased cost.

Embodiments of a silicon carbide semiconductor device and a method of manufacturing a silicon carbide semiconductor device according to the present invention will be described in detail with reference to the accompanying drawings. In the present description and accompanying drawings, layers and regions prefixed with n or p mean that majority carriers are electrons or holes. Additionally, + or - appended to n or p means that the impurity concentration is higher or lower, respectively, than layers and regions without + or -. In the description of the embodiments below and the accompanying drawings, main portions that are identical will be given the same reference numerals and will not be repeatedly described.

A structure of a silicon carbide semiconductor device according to an embodiment is described. FIG. 1 is a plan view depicting a layout when a semiconductor device according to the embodiment is viewed from a front side of a semiconductor substrate. FIG. 2 is a cross-sectional view of the structure along cutting line A-A' in FIG. 1. FIG. 3 is a table depicting examples of dimensions of intervals between adjacent FLRs in FIG. 2. A silicon carbide semiconductor device 10 according to the embodiment depicted in FIGS. 1 and 2 is a vertical MOSFET having a trench structure (device element structure) in an active region 1 of a semiconductor substrate (semiconductor chip) 40 containing silicon carbide (SiC), the silicon carbide semiconductor device 10 having a FLR structure 30 as a voltage withstanding structure in an edge termination region 2.

The active region 1 is a region in which a main current (drift current) flows when the MOSFET is ON and in a center portion 1a thereof, multiple unit cells (functional units of a device element) having a same structure are disposed adjacent to and are connected in parallel to one another. The active region 1 has a substantially rectangular shape in a plan view thereof and is disposed in substantially a center of a semiconductor substrate 40. The active region 1 is a portion in a chip center portion (a center (the chip center) portion of the semiconductor substrate 40), from a chip-outer-side end of a later-described p-type extension portion 22a (end thereof closest to an end (chip end) of the semiconductor substrate 40). The edge termination region 2 is a region between the active region 1 and the chip end, and surrounds a periphery of the active region 1 in substantially a rectangular shape. The FLR structure 30 in the edge termination region 2 is described later.

The semiconductor substrate 40 is formed by sequentially forming, by epitaxial growth, silicon carbide layers 42, 43 constituting an n⁻-type drift region 12 and a p-type base region 13, on a front surface of an n⁺-type starting substrate 41 containing silicon carbide. The semiconductor substrate 40 has a surface having the p-type silicon carbide layer 43

assumed as a front surface (first main surface) and another surface having the n⁺-type starting substrate **41** assumed as a back surface (second main surface). A portion of the p-type silicon carbide layer **43** in the edge termination region **2** is removed by etching, whereby a drop **24** is formed at the front surface of the semiconductor substrate **40**. Due to the drop **24**, the p-type silicon carbide layer **43** is left in a mesa shape in the active region **1**. The front surface of the semiconductor substrate **40** is recessed toward an n⁺-type drain region **11** to a greater extent in a portion (second surface portion) **40b** of the edge termination region **2** than in a portion (first surface portion) **40a** of the active region **1**, with the drop **24** as a boundary.

Device elements of the active region **1** and the edge termination region **2** are separated from one another by a portion (mesa edge of the drop **24**, hereinafter, the third surface portion) **40c** connecting the first surface portion **40a** and the second surface portion **40b** of the front surface of the semiconductor substrate **40**. The second surface portion **40b** of the front surface of the semiconductor substrate **40** is an exposed surface of the n⁻-type silicon carbide layer **42** exposed during formation of the drop **24**. During the formation of the drop **24**, the n⁻-type silicon carbide layer **42** may be slightly removed with the p-type silicon carbide layer **43**. The third surface portion **40c** of the front surface of the semiconductor substrate **40** is a side surface (exposed surface) of the p-type silicon carbide layer **43** exposed during the formation of the drop **24**. Being exposed at the second and the third surface portions **40b**, **40c** of the front surface of the semiconductor substrate **40** means being in contact with the interlayer insulating film **19** on the second and the third surface portions **40b**, **40c** of the front surface of the semiconductor substrate **40**.

The trench structure formed by the p-type base region (second semiconductor region) **13**, n⁺-type source regions (fourth semiconductor regions) **14**, p⁺⁺-type contact regions **15**, trenches **16**, gate insulating films **17**, and gate electrodes **18** is provided in the semiconductor substrate **40** at the first surface portion **40a** of the front surface of the semiconductor substrate **40**, in the center portion **1a** of the active region **1**. The n⁺-type starting substrate **41** is the n⁺-type drain region (first semiconductor region) **11**. The n⁻-type drift region (first semiconductor region) **12** is a portion of the n⁻-type silicon carbide layer **42** excluding later-described first and second p⁺-type regions **21**, **22**, an n-type current spreading region (not depicted), FLRs **31**, and an n⁻-type channel stopper region **32**, and is provided between the n⁺-type starting substrate **41** and these regions, the n⁻-type drift region **12** being in contact with these regions and provided spanning the active region **1** and the edge termination region **2**.

The p-type base region **13** is a portion of the p-type silicon carbide layer **43** excluding the n⁺-type source regions **14** and the p⁺⁺-type contact regions **15**. The p-type base region **13** is provided between the first surface portion **40a** of the front surface of the semiconductor substrate **40** and the n⁻-type drift region **12**. The n⁺-type source regions **14** and the p⁺⁺-type contact regions **15** are selectively provided between the first surface portion **40a** of the front surface of the semiconductor substrate **40** and the p-type base region **13**, are in contact with the p-type base region **13**, and are exposed at the first surface portion **40a** of the front surface of the semiconductor substrate **40**. Being exposed at the first surface portion **40a** of the front surface of the semiconductor substrate **40** means being in contact with a later-described source electrode **20** via contact holes of the later described interlayer insulating film **19**.

The p⁺⁺-type contact regions **15** may be omitted. In this instance, instead of the p⁺⁺-type contact regions **15**, the p-type base region **13** is exposed at the first surface portion **40a** of the front surface of the semiconductor substrate **40**. Between the n⁻-type drift region **12** and the p-type base region **13**, the n-type current spreading region (not depicted) that is a so-called current spreading layer (CSL) that reduces carrier spreading resistance may be provided at a deep position closer to the n⁺-type drain region **11** than are bottoms of the trenches **16**. Further, the first and the second p⁺-type regions (first and second high-concentration regions) **21**, **22** are provided at deep positions closer to the n⁺-type drain region **11** than are the bottoms of the trenches **16**.

The first and the second p⁺-type regions **21**, **22** have a function of mitigating electric field applied to the bottoms of the trenches **16**. The first p⁺-type regions **21** are provided apart from the p-type base region **13** and face the trenches **16** in a depth direction. The first p⁺-type regions **21** are electrically connected to the source electrode **20** by a non-depicted portion. One of the second p⁺-type regions **22** is provided between an adjacent two of the trenches **16**, apart from the first p⁺-type regions **21** and the trenches **16**, and in contact with the p-type base region **13**. The trenches **16** penetrate through the n⁺-type source regions **14** and the p-type base region **13** in the depth direction and reach the n⁻-type drift region **12** (in an instance in which the n-type current spreading region is provided, the n-type current spreading region).

The trenches **16**, for example, extend in a stripe pattern in a direction parallel to the front surface of the semiconductor substrate **40** and reach a later-described peripheral portion **1b** of the active region **1**. The p-type base region **13**, the n⁺-type source regions **14**, the p⁺⁺-type contact regions **15**, and the second p⁺-type regions **22** extend between adjacent trenches of the trenches **16** in a linear shape parallel to the trenches **16**. The p⁺⁺-type contact regions **15** may be scattered parallel to the trenches **16**. In the trenches **16**, the gate electrodes **18** are provided via the gate insulating films **17**. All the gate electrodes **18** are electrically connected via a gate runner (gate wiring layer, not depicted) of the peripheral portion **1b** of the active region **1**.

The interlayer insulating film **19** is provided on an entire area of the front surface of the semiconductor substrate **40**, the interlayer insulating film **19** covering the gate electrodes **18** in the active region **1** and covering the front surface of the semiconductor substrate **40** in the peripheral portion **1b** of the active region **1** and in the edge termination region **2**. In the peripheral portion **1b** of the active region **1** and in the edge termination region **2**, a field oxide film may be provided between the front surface of the semiconductor substrate **40** and the interlayer insulating film **19**. The source electrode (first electrode) **20**, via contact holes of the interlayer insulating film **19**, is in ohmic contact with the n⁺-type source regions **14** and the p⁺⁺-type contact regions **15** and electrically connected to the p-type base region **13**. A drain electrode (second electrode) **25** is provided in an entire area of the back surface (back surface of the n⁻-type starting substrate **41**) of the semiconductor substrate **40** and is electrically connected to the n⁺-type drain region **11**.

The peripheral portion **1b** of the active region **1** surrounds a periphery of the center portion **1a** of the active region **1** in substantially a rectangular shape. In the peripheral portion **1b** of the active region **1**, in an entire region between the first surface portion **40a** of the front surface of the semiconductor substrate **40** and the n⁻-type drift region **12**, a p-type region (second-conductivity-type peripheral region) formed by the

p⁺-type extension portion 22a, a p-type base extension portion 13a, and a p⁺⁺-type contact extension portion 15a stacked sequentially from the n⁻-type drift region 12 is provided. The p⁺-type extension portion 22a, the p-type base extension portion 13a, and the p⁺⁺-type contact extension portion 15a surround a periphery of the center portion 1a of the active region 1 in substantially a rectangular shape. The p⁺-type extension portion 22a, the p-type base extension portion 13a, and the p⁺⁺-type contact extension portion 15a each has an end facing the chip end and exposed at the third surface portion 40c of the front surface of the semiconductor substrate 40.

The p⁺-type extension portion 22a, the p-type base extension portion 13a, and the p⁺⁺-type contact extension portion 15a have a function of making electric field uniform in a plane of the first surface portion 40a of the front surface of the semiconductor substrate 40, in the peripheral portion 1b of the active region 1. The p⁺-type extension portion 22a, the p-type base extension portion 13a, and the p⁺⁺-type contact extension portion 15a are regions for leading out hole current to the source electrode 20 and have a function of suppressing concentration of the hole current during avalanche breakdown in the edge termination region 2, the hole current being generated by the n⁻-type drift region 12 in the edge termination region 2 during an OFF state of the MOSFET and flowing toward the active region 1 from the p⁺⁺-type contact extension portion 15a.

The p⁺-type extension portion 22a is an extended portion of the second p⁺-type region 22 of a unit cell closest to the chip end, of the unit cells in the center portion 1a of the active region 1. The p⁺-type extension portion 22a reaches a deep position closer to the n⁺-type drain region 11 than is the second surface portion 40b of the front surface of the semiconductor substrate 40. The p⁺-type extension portion 22a extends closer to the chip end than does the drop 24 and surrounds an entire perimeter of a boundary between the second surface portion 40b and the third surface portion 40c of the front surface of the semiconductor substrate 40. The p-type base extension portion 13a is an extended portion of the p-type base region 13. The p-type base extension portion 13a is provided between the first surface portion 40a of the front surface of the semiconductor substrate 40 and the p⁺-type extension portion 22a.

The p⁺⁺-type contact extension portion 15a is an extended portion of the p⁺⁺-type contact region 15 of the unit cell closest to the chip end, of the unit cells in the center portion 1a of the active region 1. The p⁺⁺-type contact extension portion 15a is provided between the first surface portion 40a of the front surface of the semiconductor substrate 40 and the p-type base extension portion 13a. The p⁺⁺-type contact extension portion 15a (in an instance in which the p⁺⁺-type contact extension portion 15a is omitted, the p-type base extension portion 13a) is in ohmic contact with the source electrode 20 via a contact hole of the interlayer insulating film 19. The p⁺-type extension portion 22a and the p-type base extension portion 13a are electrically connected to the source electrode 20 via the p⁺⁺-type contact extension portion 15a.

In the edge termination region 2, the n⁺-type channel stopper region 32 and multiple p⁺-type regions (FLRs: hatched portions) 31 having a floating potential and configuring the FLR structure 30 are selectively provided between the second surface portion 40b of the front surface of the semiconductor substrate 40 and the n⁻-type drift region 12. The FLR structure 30 has a function of mitigating electric field of the front side of the semiconductor substrate 40 and sustaining a breakdown voltage. The breakdown voltage is

an upper limit voltage at which no erroneous operation or destruction of the silicon carbide semiconductor device 10 occurs by an operating voltage. The FLR structure 30 is divided into at least two FLR segments (later-described FLR segments 30a, 30b, 30c) at boundaries (later-described first and second points of change b1, b2) of predetermined FLRs 31 as described hereinafter.

The FLRs 31 are provided closer to the chip end than is the p⁺-type extension portion 22a, at positions apart from the p⁺-type extension portion 22a, the FLRs 31 being provided apart from one another concentrically surrounding a periphery of the active region 1. Of the FLRs 31, the FLR 31 closest to the chip center (first FLR 31 from the chip center) is adjacent to a chip-end-side of the p⁺-type extension portion 22a in the normal direction (direction from the chip center side to the chip outer side), the chip-end-side of the p⁺-type extension portion 22a facing the chip end. All the FLRs 31 are formed in the n⁻-type silicon carbide layer 42 by ion implantation and peripheries of the FLRs 31 are surrounded by the n⁻-type drift region 12. The n⁻-type drift region 12 is between the p⁺-type extension portion 22a and the FLR 31 closest to the chip center and between all the FLRs 31 adjacent to one another. Preferably, a total number of the FLRs 31, for example, may be at least about 30. All the FLRs 31 have a same impurity concentration and a same width (width in the normal direction, i.e., a direction from the chip center to the chip end) w1.

Preferably, the impurity concentration of the FLRs 31, for example, may be at least about $1 \times 10^{18}/\text{cm}^3$ at which complete depletion does not occur even when high voltage close to the breakdown voltage of the silicon carbide semiconductor device 10 is applied to pn junctions between the FLRs 31 and the n⁻-type drift region 12 and the impurity concentration may be at most about $1 \times 10^{21}/\text{cm}^3$. When the impurity concentration of the FLRs 31 exceeds the described upper limit, the width w1 of the FLRs 31 becomes too wide due to impurity diffusion, whereby the FLRs 31 become connected to one another and thus, is undesirable. The width w1 of the FLRs 31, for example, is in a range of about 2 μm to 5 μm . In FIG. 2, a first interval between the p⁺-type extension portion 22a and the FLR 31 closest to the chip center (first FLR 31 from the chip center) is assumed as x_1 while an n-th interval between an n-th one of the FLRs 31 and an (n-1)-th one of the FLRs 31 adjacent to the n-th one and closer to the chip center than is the n-th one is assumed as x_n , and subsequent intervals in a direction from the chip center to the chip end are assumed as $x_{n+1}, \dots$ (where, $n=2$ to the total number of the FLRs 31). That is, each FLR has a first side and a second side opposite to each other, the first side is closer to the chip center than is the second side. An interval of two FLRs is a distance between one FLR and the other one that is the closest one from the second side of the one FLR.

The first interval x_1 between the p⁺-type extension portion 22a and the FLR 31 closest to the chip center, with consideration of impurity diffusion, for example, may be in a range of about 0.1 μm to 1.0 μm and preferably, for example, may be at least about 0.6 μm (refer to FIGS. 11 and 12). An n-th interval x_n between an adjacent two of the FLRs 31 is wider than the first interval x_1 between the p⁺-type extension portion 22a and the FLR 31 closest to the chip center, and an n-th interval x_n between an adjacent two of the FLRs 31 increases in arithmetic progression by a constant increase increment (width in the normal direction), the closer the adjacent two of the FLRs 31 are to the chip end, constant increase increments respectively corresponding to the FLR segments 30a to 30c. A width of the p⁺-type extension

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portion **22a** and the width w_1 of each of the FLRs **31** is wider than an opening width of an ion implantation mask by about 0.2 μm to 0.3 μm toward the chip center and toward the chip end due to impurity diffusion and therefore, an m -th interval x_m (where, $m=1$ to the total number of the FLRs **31**) is narrower than widths of portions left of the ion implantation mask (widths between openings).

An increase increment of an n -th interval x_n between an adjacent two of the FLRs **31** is constant within each FLR segment into which the FLRs **31** are divided having at least one of the FLRs **31** as a boundary, the increase increment being wider in FLR segments disposed closer to the chip end than in FLR segments disposed relatively closer to the chip center. In particular, for example, in an instance in which the increase increment of an n -th interval x_n between an adjacent two of the FLRs **31** is changed at two of the FLRs **31** set as boundaries (the first and the second points of change **b1**, **b2** sequentially from the chip center side to the chip outer side), the FLR structure **30** is divided into three FLR segments (indicated by reference characters **30a** to **30c** in a direction from the chip center side to the chip outer side). The increase increment of an n -th interval x_n between an adjacent two of the FLRs **31** is wider in the FLR segment **30b** that is between the first and the second points of change **b1**, **b2** than in the FLR segment **30a** that is closer to the chip center than is the first point of change **b1** (closest to the chip center), the FLR segment **30b** being adjacent to the FLR segment **30a** and closer to the chip end than is the FLR segment **30a**.

The increase increment of an n -th interval x_n between an adjacent two of the FLRs **31** is wider in the FLR segment **30c** than in the FLR segment **30b** between the first and the second points of change **b1**, **b2**, the FLR segment **30c** being closer to the chip end than is the second point of change **b2** that is adjacent to the FLR segment **30b** on a side thereof facing the chip end. In the respective FLR segments **30a** to **30c**, the increase increment of an n -th interval x_n between an adjacent two of the FLRs **31** is constant ($x_{n+1}-x_n=\text{constant}$). In the respective FLR segments **30a** to **30c**, an n -th interval x_n between an adjacent two of the FLRs **31** increases in arithmetic progression by the constant increase increment, the closer the adjacent two of the FLRs **31** are to the chip end. Preferably, the increase increment of an n -th interval x_n between an adjacent two of the FLRs **31**, for example, may be in a range of about 0.05 μm to 0.12 μm (refer to FIGS. 7, 8, and 11).

The first point of change **b1** (closest to the chip center) of the increase increment of an n -th interval x_n between an adjacent two of the FLRs **31** is set to a second one or a subsequent one of the FLRs **31** in the direction from the chip center side to the chip outer side, the second one being second from the chip center and assumed as a lower limit. Therefore, the FLR segment (first FLR segment) **30a** closest to the chip center includes at least a portion having the first interval x_1 between the p^+ -type extension portion **22a** and the FLR **31** closest to the chip center and a portion having a second interval x_2 between a first one of the FLRs **31** and the second one adjacent thereto of the FLRs **31**. A FLR segment (second FLR segment, here, the FLR segment **30b**) closer to the chip end than is the first point of change **b1** includes a portion having n -th intervals x_n of a third one (third interval x_3 between the second one of the FLRs **31** and an adjacent third one of the FLRs **31**) and subsequent ones in a direction from the chip center side to the chip outer side.

The point of change (closest to the chip end, here, the second point of change **b2**) of the increase increment of an n -th interval x_n between an adjacent two of the FLRs **31** is set to be a third one of the FLRs **31**, third from the chip end

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(here, for example, a 28-th one of the FLRs **31** assuming a total of 30 of the FLRs **31**) and subsequent ones in a direction from the chip outer side to the chip center side, the third FLR **31** from the chip end being assumed as an upper limit. Therefore, a FLR segment closest to the chip end (third FLR segment, here, the FLR segment **30c**) includes at least two n -th intervals x_n from the chip end (for example, 29-th and 30-th intervals x_{29} , x_{30}), and a FLR segment (second FLR segment, here, the FLR segment **30b**) closer to the chip center than is the point of change closest to the chip end includes a portion having a third one from the chip end (for example, a 28-th interval x_{29}) and subsequent n -th intervals x_n in the direction from the chip end to the chip center.

Preferably, the FLR structure **30** may be divided into at least three FLR segments. A same number of n -th intervals x_n may be included in each of the FLR segments of the FLR structure **30** (the number in the FLR segment **30a** closest to the chip center includes the first interval x_1 between the p^+ -type extension portion **22a** and the FLR **31** closest to the chip center). A reason for this is that a manner in which a depletion layer spreads in the n^- -type drift region **12**, from main junctions (pn junctions between the p -type base region **13**, the first and the second p^+ -type regions **21**, **22**, the p^+ -type extension portion **22a**, and the n^- -type drift region **12**) of the active region **1** to the chip end may be suitably set (adjusted) for a chip center side, a center, and a chip outer side of the FLR structure **30**.

When an insulating layer (field oxide film and the inter-layer insulating film **19**) covering the second surface portion **40b** of the front surface of the semiconductor substrate **40** becomes positively charged due to long periods of operation of the silicon carbide semiconductor device **10**, the spreading of the depletion layer in the n^- -type drift region **12** is suppressed due to positive charge in the insulating layer, whereby concentration of electric field in the edge termination region **2** at a portion thereof nearest the chip center is distributed by a FLR segment of the FLR structure **30** closer to the chip center (here, the FLR segment **30a**). When the insulating layer covering the second surface portion **40b** of the front surface of the semiconductor substrate **40** is negatively charged due to long periods of operation of the silicon carbide semiconductor device **10**, the spreading of the depletion layer in the n^- -type drift region **12** to the chip outer side is facilitated due to the negative charge in the insulating layer, whereby voltage decreases that occur are suppressed by a FLR segment of the FLR structure **30** closer to the chip end (here, the FLR segment **30c**).

Normally when the insulating layer covering the second surface portion **40b** of the front surface of the semiconductor substrate **40** is not charged (zero charge), spreading of the depletion layer in the n^- -type drift region **12** to the chip end may be facilitated by a FLR segment (here, the FLR segment **30b**) near a center of the FLR structure **30** and the load of the breakdown voltage near the center of the edge termination region **2** may be distributed. An example of dimensions of an m -th interval x_m described above (where, $m=1$ to the total number of the FLRs **31**) are shown in FIG. 3 as the widths of the portions left of the ion implantation mask (widths between openings) for forming the p^+ -type extension portion **22a** and the FLRs **31**. In FIG. 3, "No." indicates the number of the FLRs **31** from the chip center. In FIG. 3, the total number of the FLRs **31** is 30 and a 10-th one and a 20-th one of the FLRs **31** from the chip center are the first and the second points of change **b1**, **b2** of the increase increment for an m -th interval x_m .

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The FLR segment **30a** closest to the chip center extends between and includes the first interval x_1 between the p⁺-type extension portion **22a** and the FLR **31** closest to the chip center, and a 10-th interval x_{10} between a 9-th one and a 10-th one adjacent thereto of the FLRs **31**. The first interval x_1 , for example, is assumed to be 1 μm and in the FLR segment **30a** closest to the chip center, the increase increment of an n-th interval x_n between an adjacent two of the FLRs **31** is assumed to be 0.05 μm . In this instance, the second interval x_2 between the first one and the second one adjacent thereto of the FLRs **31** is 1.05 μm ($=1 \mu\text{m}+0.05 \mu\text{m}$), an n-th interval x_n between an adjacent two of the FLRs **31** increases in arithmetic progression by the constant increase increment of 0.05 μm , the closer the adjacent two of the FLRs **31** are to the chip end, and the 10-th interval x_{10} that is closest to the chip end in the FLR segment **30a** and between the 9-th one and the 10-th one adjacent thereto of the FLRs **31** is 1.45 μm ($=1 \mu\text{m}+0.05 \mu\text{m} \times 9$).

The FLR segment **30b** between the first and the second points of change **b1**, **b2** extends between and includes an 11-th interval x_{11} between the 10-th one and an 11-th one adjacent thereto of the FLRs **31**, and a 20-th interval x_{20} between a 19-th one and a 20-th one adjacent thereto of the FLRs **31**. The increase increment of an n-th interval x_n between an adjacent two of the FLRs **31** in the FLR segment **30b** between the first and the second points of change **b1**, **b2** is greater than that in the FLR segment **30a** closest to the chip center and is assumed to be 0.08 μm . In this instance, the 11-th interval x_{11} between the 10-th one and the 11-th one adjacent thereto of the FLRs **31** is 1.53 μm ($=1.45 \mu\text{m}+0.08 \mu\text{m}$), an n-th interval x_n between an adjacent two of the FLRs **31** increases in arithmetic progression by the constant increase increment of 0.08 μm , the closer the adjacent two of the FLRs **31** are to the chip end, and the 20-th interval x_{20} that is closest to the chip end in the FLR segment **30b** and between the 19-th one and the 20-th one adjacent thereto of the FLRs **31** is 2.25 μm ($=1.45 \mu\text{m}+0.08 \mu\text{m} \times 10$).

The FLR segment **30c** closest to the chip end extends between and includes a 21-st interval x_{21} between the 20-th one and a 21-st one adjacent thereto of the FLRs **31**, and a 30-th interval x_{30} between a 29-th one and a 30-th one adjacent thereto of the FLRs **31**. The increase increment of an n-th interval x_n between an adjacent two of the FLRs **31** is greater in the FLR segment **30c** closest to the chip end than that in the FLR segment **30b** between the first and the second points of change **b1**, **b2** and is assumed to be 0.12 μm . In this instance, the 21-st interval x_{21} between the 20-th one and the 21-st one adjacent thereto of the FLRs **31** is 2.37 μm ($=2.25 \mu\text{m}+0.12 \mu\text{m}$), an n-th interval x_n between an adjacent two of the FLRs **31** increases in arithmetic progression by the constant increase increment of 0.12 μm , the closer the adjacent two of the FLRs **31** are to the chip end, and the 30-th interval x_{30} that is closest to the chip end in the FLR segment **30c** and between the 29-th one and the 30-th one adjacent thereto of the FLRs **31** is 3.45 μm ($=2.25 \mu\text{m}+0.12 \mu\text{m} \times 10$).

Even when the insulating layer covering the second surface portion **40b** of the front surface of the semiconductor substrate **40** is charged positively or negatively, it suffices to be able to set the manner in which the depletion layer spreads in the n⁻-type drift region **12** so as to obtain breakdown voltage characteristics that are substantially the same as breakdown voltage characteristics in a normal state of the insulating layer having zero charge, and the number of the FLR segments and the number of n-th intervals x_n in each of the FLR segments of the FLR structure **30** may be

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suitably changed. Therefore, in an instance in which the total number of the FLRs **31** is 30 and the number of n-th intervals x_n in the FLR segment close to the center of the FLR structure **30** is assumed to be relatively high, for example, 20, it suffices to be able to secure the predetermined breakdown voltage of the edge termination region **2** when the insulating layer covering the second surface portion **40b** of the front surface of the semiconductor substrate **40** is not charged.

In the FLR segments respectively closest to the chip center and the chip end of the FLR structure **30**, the number of n-th intervals x_n in each is assumed to be 5, the remaining intervals. Further, of the FLR structure **30**, in the FLR segment closest to the chip center, an n-th interval x_n between an adjacent two of the FLRs **31** increases in arithmetic progression by the constant increase increment, the closer the adjacent two of the FLRs **31** are to the chip end, whereby spreading of the depletion layer in the chip center side of the edge termination region **2** to the outer side thereof may be facilitated and concentration of electric field in the chip center side of the edge termination region **2** due to positive charge in the insulating layer may be distributed. In the FLR structure **30**, in the FLR segment closest to the chip end, an interval between an adjacent two of the FLRs decreases in arithmetic progression by the constant increase increment, the closer the adjacent two FLRs are disposed to the chip center, whereby spreading of the depletion layer in the outer side of the edge termination region **2** to the chip outer side may be suppressed and breakdown voltage decreases due to negative charge in the insulating layer may be suppressed.

All the FLRs **31** are disposed at a same depth and have a same thickness. While all the FLRs **31** may reach and be exposed at the second surface portion **40b** of the front surface of the semiconductor substrate **40** (not depicted), preferably, all the FLRs **31** may be disposed deeper than the second surface portion **40b** of the front surface of the semiconductor substrate **40** (FIG. 2). In other words, a portion of the n⁻-type drift region (third semiconductor region) **12** having a predetermined thickness **t1** may be present between the second surface portion **40b** of the front surface of the semiconductor substrate **40** and all the FLRs **31**. The FLRs **31** are apart from the second surface portion **40b** of the front surface of the semiconductor substrate **40**, whereby adverse effects of charge in the insulating layer covering the second surface portion **40b** of the front surface of the semiconductor substrate **40** may be mitigated, the charge being due to long operation periods of the silicon carbide semiconductor device **10**.

The thickness **t1** of the portion of the n⁻-type drift region **12** between the second surface portion **40b** of the front surface of the semiconductor substrate **40** and the FLRs **31** may be, for example, about 0.4 μm with consideration of etching depth variation (about 0.2 μm) in forming the drop **24**. The FLRs **31** may be formed concurrently with the first p⁺-type regions **21** of the active region **1**. The FLRs **31** and the p⁺-type extension portion **22a** are disposed at the same depth, whereby electric field concentration at the chip-end-side of the p⁺-type extension portion **22a** is suppressed. Further, by forming the FLRs **31** concurrently with only the first p⁺-type regions **21**, a predetermined thickness (dimension in the depth direction) of the FLRs **31** may be stably ensured without being affected by etching depth variation in forming the drop **24**.

The n⁺-type channel stopper region **32** is apart from the FLR structure **30** and disposed closer to the chip end than is the FLR structure **30**. The n⁺-type channel stopper region **32**

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is exposed at the second surface portion **40b** of the front surface of the semiconductor substrate **40** and at the chip end. The n⁺-type channel stopper region **32** has a floating potential. The n⁺-type channel stopper region **32** surrounds a periphery of the n⁻-type drift region **12**, and the n⁻-type drift region **12** is between the n⁺-type channel stopper region **32** and the FLR **31** closest to the chip end. The n⁺-type channel stopper region **32**, for example, may have a same impurity concentration as that of the n⁺-type source regions **14**. The second surface portion **40b** of the front surface of the semiconductor substrate **40** is free of a field plate (FP) and a channel stopper electrode.

Operation of the silicon carbide semiconductor device **10** according to the embodiment is described. When voltage at least equal to a gate threshold voltage is applied to the gate electrodes **18** while voltage that is positive with respect to the source electrode **20** (forward voltage) is applied to the drain electrode **25**, a channel (n-type inversion layer) is formed along the trenches **16** of the p-type base region **13**. As a result, current flows from the n⁺-type drain region **11**, passes through the n⁻-type drift region **12** and a channel (n-type inversion layer formed along sidewalls of the trenches **16** in the p-type base region **13**) to the n⁺-type source regions **14**, and the MOSFET (the silicon carbide semiconductor device **10**) turns ON.

On the other hand, when voltage less than the gate threshold voltage is applied to the gate electrodes **18** while the forward voltage is applied between a source and drain, in the active region **1**, pn junctions (main junctions of the active region **1**) between the p-type base region **13**, the first and the second p⁺-type regions **21**, **22**, the p⁺-type extension portion **22a**, and the n⁻-type drift region **12** are reverse biased, whereby the MOSFET maintains an OFF state. Here, a depletion layer spreads in the n⁻-type drift region **12**, from the pn junctions to the n⁺-type drain region **11**, whereby electric field applied to the gate insulating films **17** at the bottoms of the trenches **16** positioned closer to the source electrode **20** than are the pn junctions is mitigated.

Further, when the MOSFET is OFF, the predetermined breakdown voltage based on breakdown field strength of silicon carbide and depletion layer width (width in the normal direction) may be ensured for an extent that the depletion layer in the n⁻-type drift region **12** extends in the edge termination region **2** to the chip outer side (chip end). In the embodiment, an n-th interval x_n between an adjacent two of the FLRs **31** of the FLR structure **30** increases in arithmetic progression by the constant increase increment, the closer the adjacent two of the FLRs **31** are to the chip end, whereby the predetermined breakdown voltage about the same as that of the conventional structure may be stably obtained even when a width w_2 of the edge termination region **2** is shorter than a width w_{102} of the edge termination region **102** of the conventional structure (FIG. **13**).

Next, a method of manufacturing the silicon carbide semiconductor device **10** according to the embodiment is described. FIGS. **4**, **5**, and **6** are cross-sectional views of states of the silicon carbide semiconductor device according to the embodiment during manufacture. In FIGS. **4** to **6**, the peripheral portion **1b** of the active region **1** of only one of multiple chip regions **50a** and the edge termination region **2** (refer to FIG. **2**) are depicted, and the center portion **1a** of the active region **1** is described with reference to FIG. **2**. Each of the chip regions **50a** is a region constituting a semiconductor chip (the semiconductor substrate **40**) after a semiconductor wafer **50** is diced (cut), the chip regions **50a** being formed in a center portion of the semiconductor wafer

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50, for example, in a matrix-like pattern, surrounded by dicing lines (cutting lines) **50b** of a grid pattern.

First, as depicted in FIG. **4**, on a front surface of an n⁺-type starting wafer **51** constituting the n⁺-type starting substrate **41**, an n⁻-type silicon carbide layer (first first-conductivity-type semiconductor layer) **42a** constituting the n⁻-type drift region **12** is epitaxially grown (first process). Next, the first p⁺-type regions **21** of the center portion **1a** of the active region **1**, lower portions (fourth portions) of the second p⁺-type regions **22** of the center portion **1a** of the active region **1**, a lower portion (first portion) **52** of the p⁺-type extension portion **22a** of the peripheral portion **1b** of the active region **1**, and all the FLRs **31** of the FLR structure **30** of the edge termination region **2** are selectively formed in surface regions of the n⁻-type silicon carbide layer **42a**, by photolithography and ion implantation of a p-type impurity, using a single ion implantation mask (second process).

The FLRs **31** of the FLR structure **30** are formed concurrently with the first p⁺-type regions **21**, whereby a separate ion implantation process for forming the FLRs **31** becomes unnecessary and the number of processes may be reduced. Further, other regions (the first p⁺-type regions **21**, lower portions of the second p⁺-type regions **22**, and the p⁺-type extension portion **22a**) and the FLRs **31** are formed concurrently by the same ion implantation mask, whereby the number of ion implantation masks may be reduced and manufacturing cost may be reduced. Further, the FLRs **31** of the FLR structure **30** are formed concurrently with the lower portion **52** of the p⁺-type extension portion **22a**, whereby the FLRs **31** may be disposed at the same depth as that of the p⁺-type extension portion **22a**.

Next, as depicted in FIG. **5**, the ion implantation mask (not depicted) used in forming the first p⁺-type regions **21**, etc. is removed and thereafter, an n⁻-type silicon carbide layer (second first-conductivity-type semiconductor layer) **42b** is further epitaxially grown on the n⁻-type silicon carbide layer **42a**, thereby increasing the thickness (third process), whereby the n⁻-type silicon carbide layer **42** (**42a**, **42b**) having a product (the silicon carbide semiconductor device **10**) thickness is formed. Next, in the center portion **1a** of the active region **1**, upper portions (fifth portions) of the second p⁺-type regions **22** are formed in the n⁻-type silicon carbide layer **42b**, by photolithography and ion implantation of a p-type impurity. The upper portions and the lower portions of the second p⁺-type regions **22** are continuous in the depth direction, whereby the second p⁺-type regions **22** are formed.

In the peripheral portion **1b** of the active region **1**, an upper portion (second portion) **53** of the p⁺-type extension portion **22a** is formed in the n⁻-type silicon carbide layer **42b** concurrently with the upper portions of the second p⁺-type regions **22** (fourth process). The upper portion **53** and the lower portion **52** of the p⁺-type extension portion **22a** are continuous in the depth direction, whereby the p⁺-type extension portion **22a** is formed. Ion implantation is not performed in the n⁻-type silicon carbide layer **42b** in the edge termination region **2**. Therefore, in the edge termination region **2**, all the FLRs **31** are covered by the n⁻-type silicon carbide layer **42b** left as the n⁻-type drift region **12**. Next, at the surface of the n⁻-type silicon carbide layer **42**, the p-type silicon carbide layer **43** constituting the p-type base region **13** is epitaxially grown (fifth process).

By the processes up to here, the semiconductor wafer **50** in which the silicon carbide layers **42**, **43** constituting the n⁻-type drift region **12** and the p-type base region **13** are epitaxially grown on the front surface of the n⁺-type starting wafer **51** is completed. In an instance in which the n-type

current spreading region (not depicted) is formed, in an entire area of the active region **1**, lower and upper portions of the n-type current spreading region are respectively formed in the n⁻-type silicon carbide layers **42a**, **42b** so as to be continuous in the depth direction, by photolithography and ion implantation of an n-type impurity performed for each epitaxial growth of the n⁻-type silicon carbide layers **42a**, **42b** constituting the n⁻-type drift region **12**.

Next, as depicted in FIG. 6, by photolithography and etching, a portion of the p-type silicon carbide layer **43** in the edge termination region **2** is removed and the p-type silicon carbide layer **43** is left only in the active region **1**. As a result, at the front surface of the semiconductor wafer **50**, the drop **24** where a portion closer to the chip end (the second surface portion **40b**) than is a portion closer to the chip center is lower (recessed) toward the n⁺-type starting wafer **51** is formed and in the edge termination region **2**, the n⁻-type silicon carbide layer **42b** is exposed at the second surface portion **40b** of the front surface of the semiconductor wafer **50**. A surface region of the n⁻-type silicon carbide layer **42b** exposed at the second surface portion **40b** of the front surface of the semiconductor wafer **50** may be slightly removed.

For example, in an instance in which the FLRs **31** are formed in the n⁻-type silicon carbide layer **42b**, a surface region of the n⁻-type silicon carbide layer **42b** is slightly removed by the etching for forming the drop **24**, whereby the thickness of the FLRs **31** changes. On the other hand, as described above, the FLRs **31** are formed only in the n⁻-type silicon carbide layer **42a** and the n⁻-type silicon carbide layer **42b** is free of the FLRs **31**, therefore, even when a surface region of the n⁻-type silicon carbide layer **42b** is slightly removed during formation of the drop **24**, the FLRs **31** may be left having a predetermined thickness. Further, the n⁻-type silicon carbide layer **42b** constituting the n⁻-type drift region **12** may be left on the FLRs **31**.

Next, the etching mask (not depicted) used to partially remove the p-type silicon carbide layer **43** is removed. Next, a process including photolithography, ion implantation, and ion implantation mask (not depicted) removal as one set is repeatedly performed under different conditions, whereby in surface regions of the semiconductor wafer **50**, at the front surface thereof (main surface having the p-type silicon carbide layer **43**), the n⁺-type source regions **14**, the p⁺⁺-type contact regions **15**, and the p⁺⁺-type contact extension portion **15a** are selectively formed in the p-type silicon carbide layer **43**. A sequence in which the n⁺-type source regions **14** and the p⁺⁺-type contact regions **15** are formed may be interchanged.

Concurrently with the formation of the n⁺-type source regions **14**, in a surface region (surface region of the n⁻-type silicon carbide layer **42**) of the semiconductor wafer **50**, at the second surface portion **40b** of the front surface thereof, the n⁺-type channel stopper region **32** may be selectively formed straddling between ends of the chip regions **50a** adjacent to one another. A portion of the n⁻-type silicon carbide layer **42** (**42a**, **42b**) excluding the first and the second p⁺-type regions **21**, **22**, the p⁺-type extension portion **22a**, the FLRs **31**, and the n⁺-type channel stopper region **32** constitutes the n⁻-type drift region **12**. A portion of the p-type silicon carbide layer **43** excluding the n⁺-type source regions **14**, the p⁺⁺-type contact regions **15**, and the p⁺⁺-type contact extension portion **15a** constitutes the p-type base region **13** and the p-type base extension portion (third portion) **13a**.

Next, ion-implanted impurities are activated by a heat treatment. Next, by a general method, the trenches **16**, the

gate insulating films **17**, the gate electrodes **18**, the interlayer insulating film **19**, the source electrode **20** (sixth process), the drain electrode **25** (seventh process), and a passivation film (polyimide protective film, not depicted) are formed. Next, portions of the passivation film on the dicing lines **50b** are removed. Thereafter, the semiconductor wafer **50** is diced along the dicing lines **50b**, whereby the chip regions **50a** are formed into individual semiconductor chips (the semiconductor substrate **40**) and the silicon carbide semiconductor device **10** in FIGS. **1** and **2** is completed.

As described above, according to the embodiment, the FLR structure configured by multiple FLRs each having a floating potential surrounding a periphery of the active regions is provided in the edge termination region. The FLR structure is divided into at least two FLR segments with a predetermined FLR as a boundary. The interval between an adjacent two of the FLRs increases in arithmetic progression by a constant increase increment, the closer the adjacent two FLRs are to the chip end, the constant increase increments respectively corresponding to the FLR segments, and the increase increment for a corresponding FLR segment is greater, the closer the FLR segment is to the chip end. As a result, the length of the edge termination region may be reduced, margins for dimension variation of the ion implantation mask may be taken, and variation of the breakdown voltage due to charge in the insulating layer on the surface of the semiconductor substrate in the edge termination region may be suppressed, the charge being due to long periods of operation.

Further, according to the embodiment, the length of the edge termination region is reduced, whereby increases in material cost may be suppressed. Further, according to the embodiment, the voltage withstanding structure is assumed to be a FLR structure, whereby the voltage withstanding structure may be formed by a single ion implantation process using one ion implantation mask, the number of masks and the number of processes may be reduced compared to an instance in which the voltage withstanding structure is a JTE structure, and manufacturing cost may be suppressed. Further, the FLRs are formed concurrently with the first p-type regions of the active region, whereby the number of masks and the number of processes may be further reduced. Therefore, a low-cost silicon carbide semiconductor device having a voltage withstanding structure capable of stably ensuring the predetermined breakdown voltage and formed by fewer processes may be provided.

Breakdown voltage characteristics of the silicon carbide semiconductor device **10** according to the embodiment described above (hereinafter, first and second examined examples, refer to FIG. **2**) were verified. FIGS. **7**, **8**, **9**, and **10** are characteristics diagrams depicting results of simulation of breakdown voltage characteristics of a conventional example. FIGS. **11** and **12** are characteristics diagrams depicting results of simulation of breakdown voltage characteristics of the first and the second examined examples, respectively. In the first examined example, the FLR structure **30** is divided into the FLR segments **30a** to **30c** under the dimension conditions in FIG. **3** described above. In the second examined example, the total number of the FLRs **31** is different from that in the first examined example. In the first and the second examined examples, the impurity concentration of the FLRs **31** and the width **w1** were assumed to be $1 \times 10^{18}/\text{cm}^3$ and $3 \mu\text{m}$, respectively. The thickness **t1** of the portion of the n⁻-type drift region **12** between the second surface portion **40b** of the front surface of the semiconductor substrate **40** and the FLRs **31** was assumed to be $0.2 \mu\text{m}$.

For comparison, reliability of the breakdown voltage of the edge termination region **102** of the conventional silicon carbide semiconductor device **110** (hereinafter, the conventional example, refer to FIG. **13**) was verified. The conventional example differs from the first examined example in that the general FLR structure **130** is provided in the edge termination region **102**. Accordingly, in the conventional example, the FLR structure **130** is not divided into FLR segments, the increase increment of an i -th interval x_j between an adjacent two of the FLRs **131** is constant in the entire region of the FLR structure **130** (where, $i=2$ to the total number of the FLRs **131**, $j=i+100$). In the conventional example, the impurity concentration of the FLRs **131**, the width w_{101} of the FLRs **131**, and the thickness t_{101} of the n^- -type drift region **112** between the second surface portion **140b** of the front surface of the semiconductor substrate **140** and the FLRs **131** were the same as those in the first examined example.

First, breakdown voltage characteristics of the edge termination region **102** of the conventional example are described. The increase increment (horizontal axis) of an i -th interval x_j between an adjacent two of the FLRs **131** of the FLR structure **130** of the conventional example was variously changed, and results of simulation of a breakdown voltage BV_{dss} (vertical axis) of the edge termination region **102** are shown in FIGS. **7** and **8**. FIGS. **7** and **8** respectively show instances in which the first interval x_{101} between the p^+ -type extension portion **122a** and the FLR **131** closest to the chip center was assumed to be $1.0\ \mu\text{m}$ and $0.7\ \mu\text{m}$. In the conventional example in FIGS. **7** and **8**, the total number of the FLRs **131** was assumed to be 30. FIGS. **7** and **8** show an instance in which the insulating layer (field oxide film and the interlayer insulating film **119**) covering the second surface portion **140b** of the front surface of the semiconductor substrate **140** is charged positively (positive charge is accumulated), an instance in which the insulating layer is negatively charged (negative charge is accumulated), and a normal state when the insulating layer is not charged (zero charge) (similarly in FIGS. **9** and **10**).

From the results in FIGS. **7** and **8**, in the conventional example, breakdown voltage variation was confirmed to occur due to charge accumulated in the insulating layer covering the second surface portion **140b** of the front surface of the semiconductor substrate **140** (hereinafter, simply “insulating layer”), the accumulated charge being due to long periods of operation under a high temperature. In particular, in a setting in which the increase increment of an i -th interval x_j between an adjacent two of the FLRs **131** is reduced (origin-side of horizontal axis), when the insulating layer is negatively charge, the breakdown voltage tends to decrease compared to the normal state. In a setting in which the increase increment of an i -th interval x_j between an adjacent two of the FLRs **131** is increased (side apart from origin of horizontal axis), when the insulating layer is positively charged, the breakdown voltage tends to decrease compared to the normal state. Further, for a setting in which the increase increment of an i -th interval x_j between an adjacent two of the FLRs **131** is assumed to be $0.075\ \mu\text{m}$, while the breakdown voltage characteristics are the most stable, for all the settings in FIGS. **7** and **8**, compared to the normal state, breakdown voltage variation of at least 100V occurred.

Thus, results of simulation of the breakdown voltage (vertical axis) of the edge termination region **102** by assuming the increase increment of an i -th interval x_j between an adjacent two of the FLRs **131** to be $0.075\ \mu\text{m}$ and variously changing the first interval x_{101} (horizontal axis) between the

p^+ -type extension portion **122a** and the FLR **131** closest to the chip center are shown in FIGS. **9** and **10**. In the conventional example in FIGS. **9** and **10**, the total number of the FLRs **131** was assumed to be 30 and 60, respectively. In FIGS. **9** and **10**, a horizontal axis indicates widths of the portions left of the ion implantation mask (width covering portion corresponding to the first interval x_{101}) for forming the first interval x_{101} between the p^+ -type extension portion **122a** and the FLR **131** closest to the chip center. A mask-dimension lower limit is a lower limit of a necessary width of the portions left of the ion implantation mask so that the first interval x_{101} between the p^+ -type extension portion **122a** and the FLR **131** closest to the chip center is not lost due to impurity diffusion.

From the results in FIG. **9**, it was confirmed that breakdown voltage characteristics are stable only at one point, a setting $c2$ that is the same as a setting $c1$ (setting in which the increase increment of an i -th interval x_j between an adjacent two of the FLRs **131** is assumed to be $0.075\ \mu\text{m}$ and the first interval x_{101} between the p^+ -type extension portion **122a** and the FLR **131** closest to the chip center is assumed to $0.7\ \mu\text{m}$) in FIG. **8** for which breakdown voltage characteristics were the most stable of the settings of the conventional example in FIGS. **7** and **8**. It was confirmed that in a setting in which the first interval x_{101} between the p^+ -type extension portion **122a** and the FLR **131** closest to the chip center is reduced (origin-side of horizontal axis), breakdown voltage variation due to negative charge increases and in a setting in which the first interval x_{101} between the p^+ -type extension portion **122a** and the FLR **131** closest to the chip center is increased (side apart from origin of horizontal axis), breakdown voltage variation due to positive charge increases.

From the results in FIG. **10**, it was confirmed that by increasing the total number of the FLRs **131** to 60 to twice that in the conventional example in FIG. **9**, in the setting in which the first interval x_{101} between the p^+ -type extension portion **122a** and the FLR **131** closest to the chip center is reduced, while breakdown voltage variation due to negative charge may be suppressed, in a setting in which the first interval x_{101} between the p^+ -type extension portion **122a** and the FLR **131** closest to the chip center is increased, breakdown voltage variation due to positive charge is not improved. Further, the inventors confirmed that the narrower is the first interval x_{101} between the p^+ -type extension portion **122a** and the FLR **131** closest to the chip center, the more difficult the ion implantation process of the FLRs **131** becomes. Further, it was confirmed that the width w_{102} of the edge termination region **102** of the conventional example in FIG. **10** is $355\ \mu\text{m}$, which is at least two times longer than the width w_{102} of the edge termination region **102** ($=144\ \mu\text{m}$) of the conventional example in FIG. **9**.

On the other hand, from the results in FIGS. **11** and **12**, it was confirmed that in the first and the second examined examples, breakdown voltage variation due to charge in the insulating layer (field oxide film and the interlayer insulating film **19**) covering the second surface portion **40b** of the front surface of the semiconductor substrate **40** is suppressed compared to the conventional example and compared to the normal state, is less than 100V, the charge being due to long periods of operation under a high temperature. In the first and the second examined examples, the first interval x_1 (horizontal axis) between the p^+ -type extension portion **22a** and the FLR **31** closest to the chip center was variously changed and results of simulation of the breakdown voltage BV_{dss} (vertical axis) of the edge termination region **2** are shown in FIGS. **11** and **12**. FIGS. **11** and **12** shown an

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instance in which the insulating layer covering the second surface portion **40b** of the front surface of the semiconductor substrate **40** (hereinafter, simply “insulating layer”) is positively charged (positive charge is accumulated), an instance in which the insulating layer is negatively charged (negative charge is accumulated), and a normal state when the insulating layer is not charged (zero charge).

In FIGS. **11** and **12**, a horizontal axis indicates the widths of the portions left of the ion implantation mask (width covering the first interval x_1) for forming the first interval x_1 between the p⁺-type extension portion **22a** and the FLR **31** closest to the chip center. A mask-dimension lower limit is a lower limit of a necessary width of the portions left of the ion implantation mask so that the first interval x_1 between the p⁺-type extension portion **22a** and the FLR **31** closest to the chip center is not lost due to impurity diffusion. In particular, it was confirmed that in the first and the second examined examples (FIGS. **11** and **12**), by setting the first interval x_1 between the p⁺-type extension portion **22a** and the FLR **31** closest to the chip center to be at most 1 μm , breakdown voltage variation due to charge in the insulating layer is suppressed, and stable breakdown voltage characteristics compared to the conventional examples in FIGS. **9** and **10** may be obtained.

Therefore, the first interval x_1 between the p⁺-type extension portion **22a** and the FLR **31** closest to the chip center is set to be at least 1 μm , whereby margins for dimension variation of the ion implantation mask for forming the FLRs **31** may be taken. Further, from the results for the second examined example in FIG. **12**, it was confirmed that by increasing the total number of the FLRs **31**, breakdown voltage characteristics may be further stabilized and in a setting in which the first interval x_1 between the p⁺-type extension portion **22a** and the FLR **31** closest to the chip center is in a range of 0.6 μm to 1.0 μm , breakdown voltage variation is nearly absent. In the second examined example, it was assumed that 12 of the FLRs **31** are disposed in each of the three FLR segments **30a** to **30c** and a k-th interval x_k between an adjacent two of the FLRs **31** is a dimension different from that in the first examined example ($k=2$ to 36).

Further, in the conventional example (refer to FIG. **10**), in contrast to breakdown voltage characteristics not being stable even when a total number of the FLRs **131** is set to 60 and the width w_{102} of the edge termination region **102** is increased to 355 μm , in the second examined example, stable breakdown voltage characteristics may be obtained even when the width w_2 of the edge termination region **2** is reduced to 171 μm .

In the foregoing, the present invention is not limited to the embodiments described above and various changes within a range not departing from the spirit of the invention are possible. For example, the present invention is suitably applicable even in an instance in which the front surface of the semiconductor substrate spanning the active region and the edge termination region is flat (free of the drop). Further, the present invention is similarly implemented when conductivity types (n-type, p-type) are reversed.

According to the present invention, the voltage withstanding structure is the FLR structure, whereby the voltage withstanding structure may be formed by a single session of ion implantation and therefore, compared to an instance in which the voltage withstanding structure is a JTE structure, the number of masks and the number of processes may be reduced and manufacturing cost may be reduced. Further, according to the invention described above, the length of the termination region may be reduced, margins for dimension variation of the ion implantation mask may be taken, and

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breakdown voltage variation due to charge in the insulating layer on the front surface of semiconductor substrate in the termination region may be reduced, the charge being due to operation for long periods.

The present invention achieves an effect in that a low-cost silicon carbide semiconductor device having a voltage withstanding structure capable of stably ensuring a predetermined breakdown voltage and formed by few processes, and a method of manufacturing the silicon carbide semiconductor device may be provided.

As described above, the silicon carbide semiconductor device and the method of manufacturing a silicon carbide semiconductor device according to the present invention are useful for power semiconductor devices used in power converting equipment, power source devices of various types of industrial machines, etc.

Although the invention has been described with respect to a specific embodiment for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art which fairly fall within the basic teaching herein set forth.

What is claimed is:

1. A silicon carbide semiconductor device, comprising:
 - a semiconductor substrate containing silicon carbide and having an active region and a termination region surrounding a periphery of the active region, the semiconductor substrate having a first main surface and a second main surface opposite to each other;
 - a first semiconductor region of a first conductivity type, provided in the semiconductor substrate, spanning the active region and the termination region;
 - a second semiconductor region of a second conductivity type, provided between the first main surface of the semiconductor substrate and the first semiconductor region in the active region;
 - a device element structure having a pn junction between the first semiconductor region and the second semiconductor region, a current that energizes the pn junction flows in the device element structure;
 - a second-conductivity-type peripheral region surrounding a periphery of the device element structure, provided between the first main surface of the semiconductor substrate and the first semiconductor region, between the device element structure and the termination region;
 - a first electrode provided on the first main surface of the semiconductor substrate, electrically connected to the second semiconductor region and the second-conductivity-type peripheral region;
 - a second electrode provided on the second main surface of the semiconductor substrate, electrically connected to the first semiconductor region;
 - a plurality of field limiting rings (FLRs) of the second conductivity type, provided at the first main surface of the semiconductor substrate so as to form portions of the first main surface, the FLRs having a floating potential and configuring a FLR structure, the FLRs being provided apart from one another with respective intervals therebetween, each interval being a shortest distance between an adjacent two of the FLRs that are adjacent to each other, and each FLR concentrically surrounding the periphery of the active region, the plurality of FLRs being provided between the first main surface of the semiconductor substrate and the first semiconductor region in the termination region and respectively having a same width in a radial direction thereof that is parallel to the first main surface of the

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semiconductor substrate, the plurality of FLRs each facing a surface of the second-conductivity-type peripheral region, the surface of the second-conductivity-type peripheral region facing the termination region in a direction parallel to the first main surface of the semiconductor substrate; and

an insulating layer provided on the first main surface of the semiconductor substrate, the insulating layer covering surfaces of the plurality of FLRs entirely in the termination region, wherein

the plurality of FLRs includes at least one boundary FLR, and is divided into at least two FLR segments with the at least one boundary FLR as a boundary, each FLR segment including at least 12 FLRs,

an interval between any adjacent two of the FLRs that are adjacent to each other is wider than an interval between the second-conductivity-type peripheral region and a first FLR of the plurality of FLRs closest to the second-conductivity-type peripheral region,

within a same FLR segment, an interval of an adjacent two of the FLRs increases as the adjacent two of the FLRs approaches an end of the semiconductor substrate by an increment in arithmetic progression,

each increment in arithmetic progression of each FLR segment among the at least two FLR segments increases as said each FLR segment approaches the end of the semiconductor substrate, resulting in different increments in arithmetic progression respectively in respective ones of the at least two FLR segments, and the interval between the second-conductivity-type peripheral region and the first FLR closest to the second-conductivity-type peripheral region is in a range of 0.6 μm to 1.0 μm .

2. The silicon carbide semiconductor device according to claim 1, further comprising

a third semiconductor region of the first conductivity type, provided between the first main surface of the semiconductor substrate and the plurality of FLRs.

3. The silicon carbide semiconductor device according to claim 1, wherein

a total number of FLRs included in the plurality of FLRs is at least 30.

4. The silicon carbide semiconductor device according to claim 1, wherein

each FLR of the plurality of FLRs has an impurity concentration in a range of $1 \times 10^{18}/\text{cm}^3$ to $1 \times 10^{21}/\text{cm}^3$.

5. The silicon carbide semiconductor device according to claim 1, wherein said same width of each FLR of the plurality of FLRs in a range of 2 μm to 5 μm .

6. The silicon carbide semiconductor device according to claim 2, wherein

the third semiconductor region has a thickness that is at most 0.4 μm .

7. The silicon carbide semiconductor device according to claim 1, wherein

said each increment is in a range of 0.05 μm to 0.12 μm .

8. The silicon carbide semiconductor device according to claim 1, wherein

the at least two FLR segments includes a first FLR segment that is closest to the second-conductivity-type peripheral region and a second FLR segment that is adjacent to an end-side of the first FLR segment, the end-side of first FLR segment facing a chip end, the boundary between the first FLR segment and the second FLR segment being located between a second or subsequent FLR of the plurality of FLRs from the second-conductivity-type peripheral region and a FLR

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of the plurality of FLRs closer to the second-conductivity-type peripheral region than is the second or subsequent FLR.

9. The silicon carbide semiconductor device according to claim 8, wherein

the at least two FLR segments further includes a third FLR segment closest to the chip end and the second FLR segment adjacent to a side of the third FLR segment, the side of the third FLR segment facing the second-conductivity-type peripheral region, the boundary between the third FLR segment and the second FLR segment being located between one FLR among the plurality of FLRs closer to the second-conductivity-type peripheral region than is a third or subsequent FLR from the chip end, and another FLR among the plurality of FLRs closer to the second-conductivity-type peripheral region than is the one FLR.

10. The silicon carbide semiconductor device according to claim 1, wherein

the second-conductivity-type peripheral region has an upper portion and a lower portion, the upper portion being relatively closer to the first main surface of the semiconductor substrate and having an impurity concentration that is a same as an impurity concentration of the second semiconductor region, the lower portion being relatively closer to the first semiconductor region and having an impurity concentration that is the same as an impurity concentration of the FLRs.

11. The silicon carbide semiconductor device according to claim 1,

wherein the device element structure includes:

a plurality of fourth semiconductor regions of the first conductivity type, selectively provided between the first main surface of the semiconductor substrate and the second semiconductor region, the fourth semiconductor regions being electrically connected to the first electrode;

a plurality of trenches penetrating through the fourth semiconductor regions and the second semiconductor region and reaching the first semiconductor region;

a plurality of gate electrodes provided in the trenches via a plurality of gate insulating films;

a plurality of first high-concentration regions of the second conductivity type, selectively provided between the first semiconductor region and the second semiconductor region, apart from the second semiconductor region, and closer to the second electrode than are bottoms of the trenches, the first high-concentration regions facing the bottoms of the trenches in a depth direction and having an impurity concentration higher than an impurity concentration of the second semiconductor region; and

a plurality of second high-concentration regions of the second conductivity type, selectively provided between the first semiconductor region and the second semiconductor region, apart from the trenches and the first high-concentration regions, and in contact with the second semiconductor region, the second high-concentration regions being closer to the second electrode than are the bottoms of the trenches and having a same impurity concentration as the impurity concentration of the first high-concentration regions, and

wherein the plurality of FLRs and the first high-concentration regions have impurity concentrations that are the same as each other.

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12. A method of manufacturing the silicon carbide semiconductor device according to claim 1, the method comprising:

- forming a first first-conductivity-type semiconductor layer constituting the first semiconductor region; 5
- selectively forming a first portion of the second-conductivity-type peripheral region and the plurality of FLRs in surface regions of the first first-conductivity-type semiconductor layer;
- forming a second first-conductivity-type semiconductor layer constituting the first semiconductor region, on the first first-conductivity-type semiconductor layer; 10
- selectively forming a second portion of the second-conductivity-type peripheral region at a position facing the first portion of the second first-conductivity-type semiconductor layer in a depth direction, the second portion reaching the first portion; 15
- forming a second-conductivity-type semiconductor layer on the second first-conductivity-type semiconductor layer in the active region, a portion of the second-conductivity-type semiconductor layer facing the second portion in the depth direction constituting a third portion of the second-conductivity-type peripheral region and a remaining portion constituting the second semiconductor region; 20
- forming the insulating layer on the first main surface of the semiconductor substrate so as to cover the surfaces of the plurality FLRs entirely in the termination region; 25
- forming the first electrode to be electrically connected to the second semiconductor region and the second-conductivity-type peripheral region; and 30
- forming the second electrode to be electrically connected to the first semiconductor region. 35

13. The method according to claim 12,

wherein the device element structure includes:

- a plurality of fourth semiconductor regions of the first conductivity type, selectively provided between the first main surface of the semiconductor substrate and the second semiconductor region, the fourth semiconductor regions being electrically connected to the first electrode; 35
- a plurality of trenches penetrating through the fourth semiconductor regions and the second semiconductor region and reaching the first semiconductor region; 40

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a plurality of gate electrodes provided in the trenches via a plurality of gate insulating films;

a plurality of first high-concentration regions of the second conductivity type, selectively provided between the first semiconductor region and the second semiconductor region, apart from the second semiconductor region, and closer to the second electrode than are bottoms of the trenches, the first high-concentration regions facing the bottoms of the trenches in the depth direction and having an impurity concentration higher than an impurity concentration of the second semiconductor region; and

a plurality of second high-concentration regions of the second conductivity type, selectively provided between the first semiconductor region and the second semiconductor region, apart from the trenches and the first high-concentration regions, and in contact with the second semiconductor region, the second high-concentration regions being closer to the second electrode than are the bottoms of the trenches and having a same impurity concentration as the impurity concentration of the first high-concentration regions,

wherein said selectively forming the first portion and the plurality of FLRs further includes selectively forming the first high-concentration regions and respective fourth portions of the second high-concentration regions in the surface regions of the first first-conductivity-type semiconductor layer, and

wherein said selectively forming the second portion includes

in the second first-conductivity-type semiconductor layer, selectively forming the second portion reaching the first portion at a position facing the first portion in the depth direction, and

selectively forming respective fifth portions of the second high-concentration regions reaching the respective fourth portions at positions respectively facing the respective fourth portions in the depth direction.

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